FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

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1. Motivace

1.1 Dobový kontext

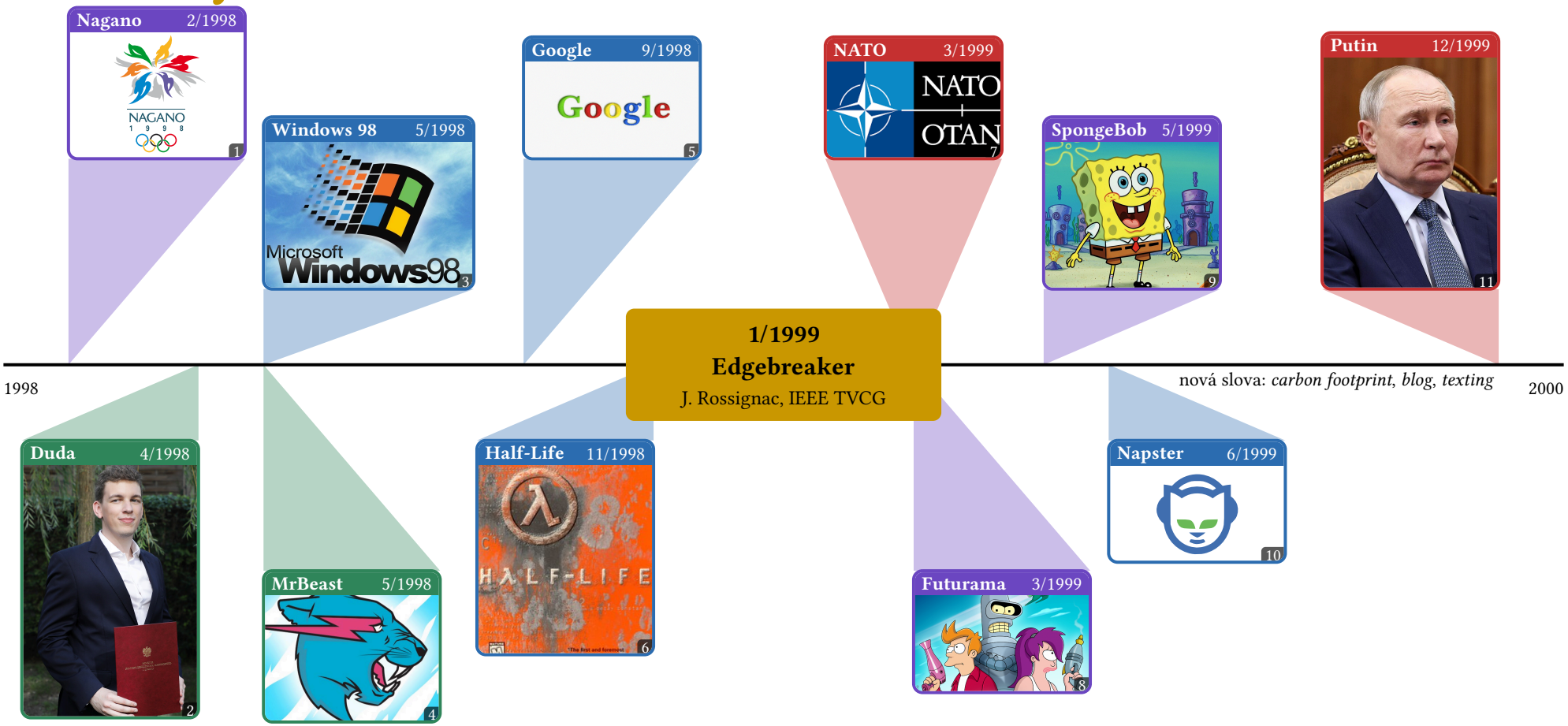


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext



Figure 2: Jarek Rossignac, zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

2. Definice problému

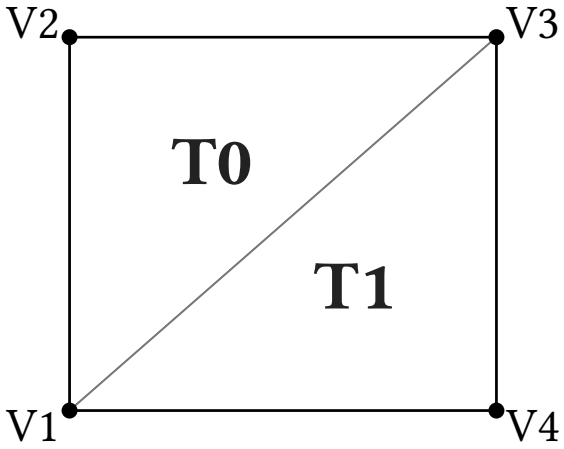


Figure 3: Trojúhelníková mesh.

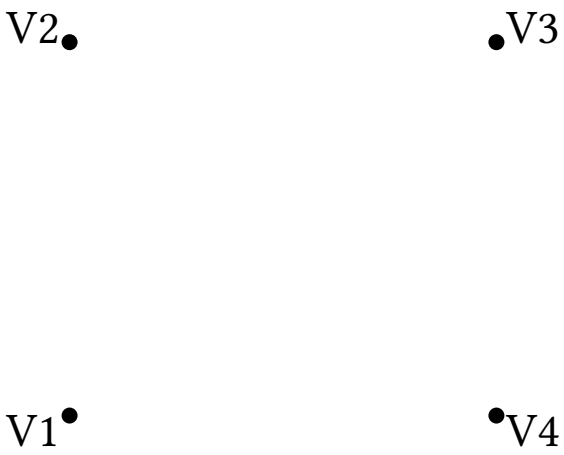


Figure 3: Trojúhelníková mesh.

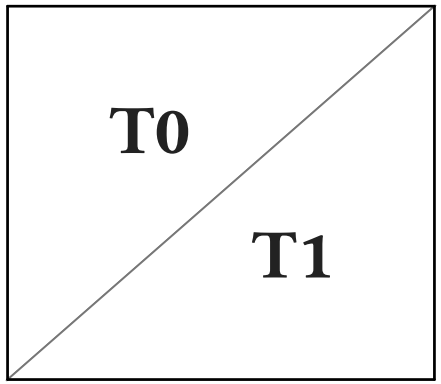


Figure 3: Trojúhelníková mesh.

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]

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[V1 V2 V3]
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[V3 V1 V2]
[V2 V3 V1]

2.2 Jak schovat informaci?

pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

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pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

CLERS řetězec

2.2 Jak schovat informaci?

pořadí vertexů	CLERS řetězec
[V1 V2 V3]	CLRSLRCLEE
[V1 V3 V2]	
[V2 V1 V3]	
[V3 V1 V2]	
[V2 V3 V1]	
[V3 V2 V1]	

3. Komprese

3.1 Pravidla

History =

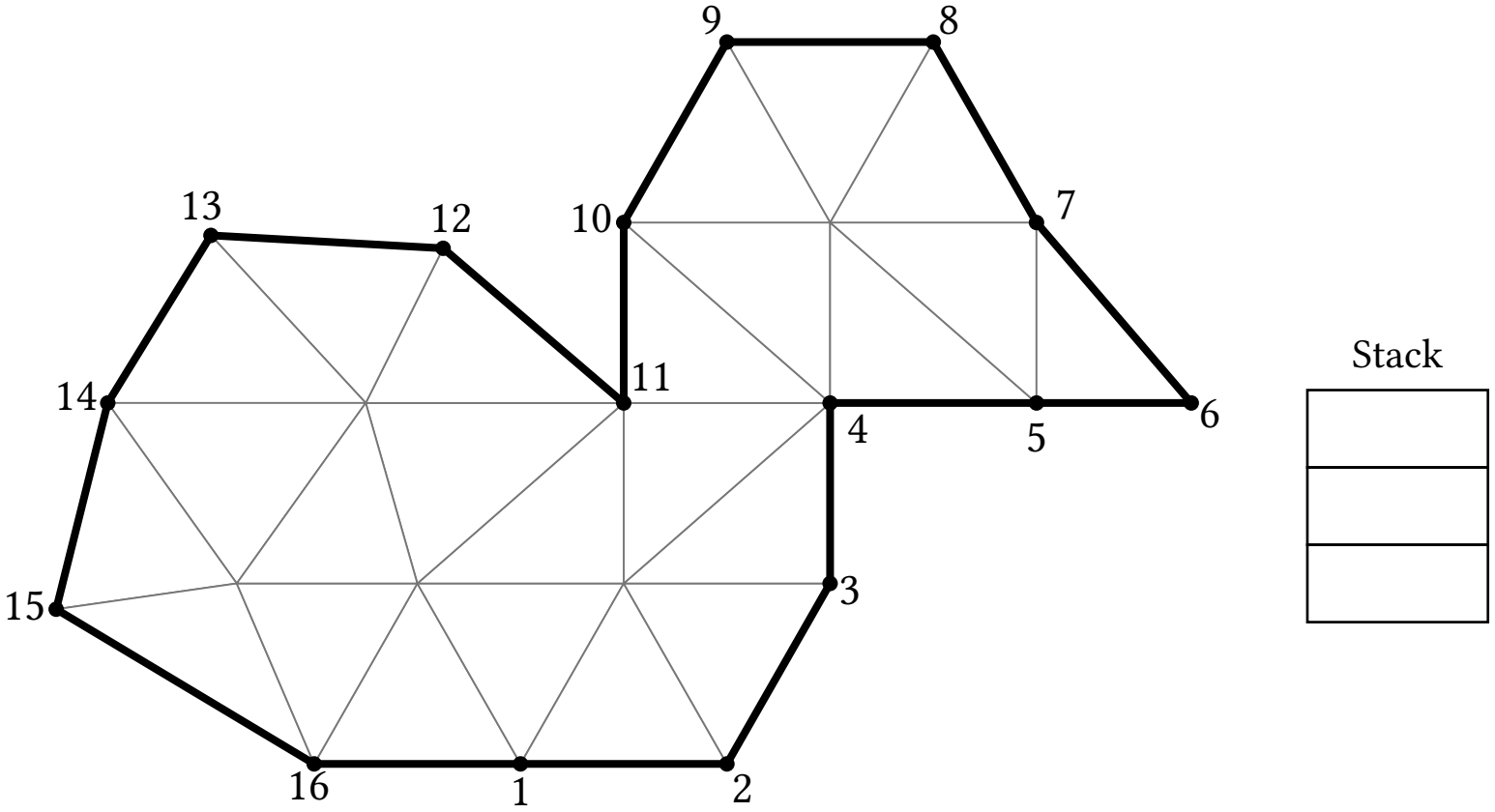


Figure 4: Jednoduchá komprese.

History = C

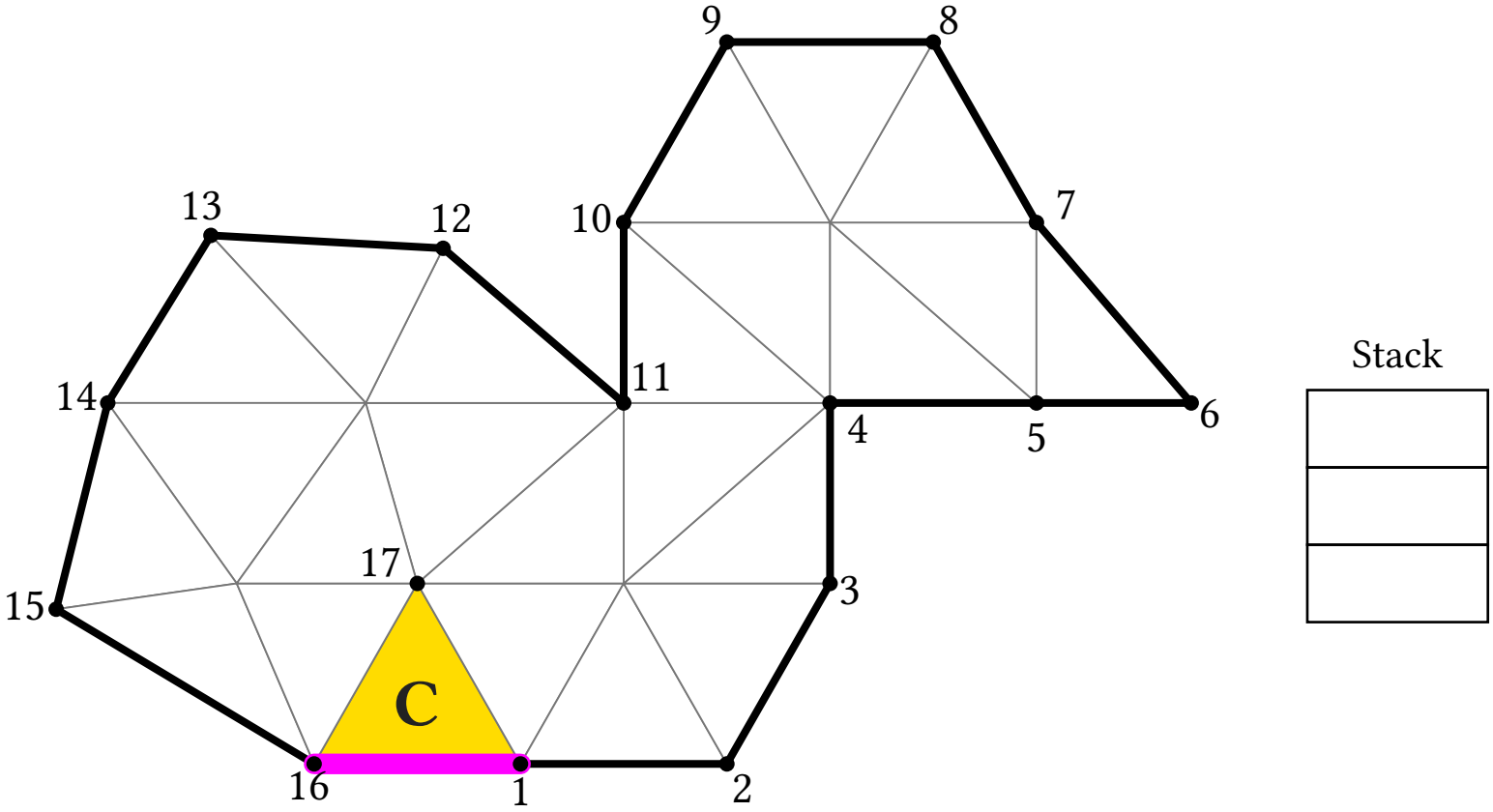


Figure 4: Jednoduchá komprese.

History = CC

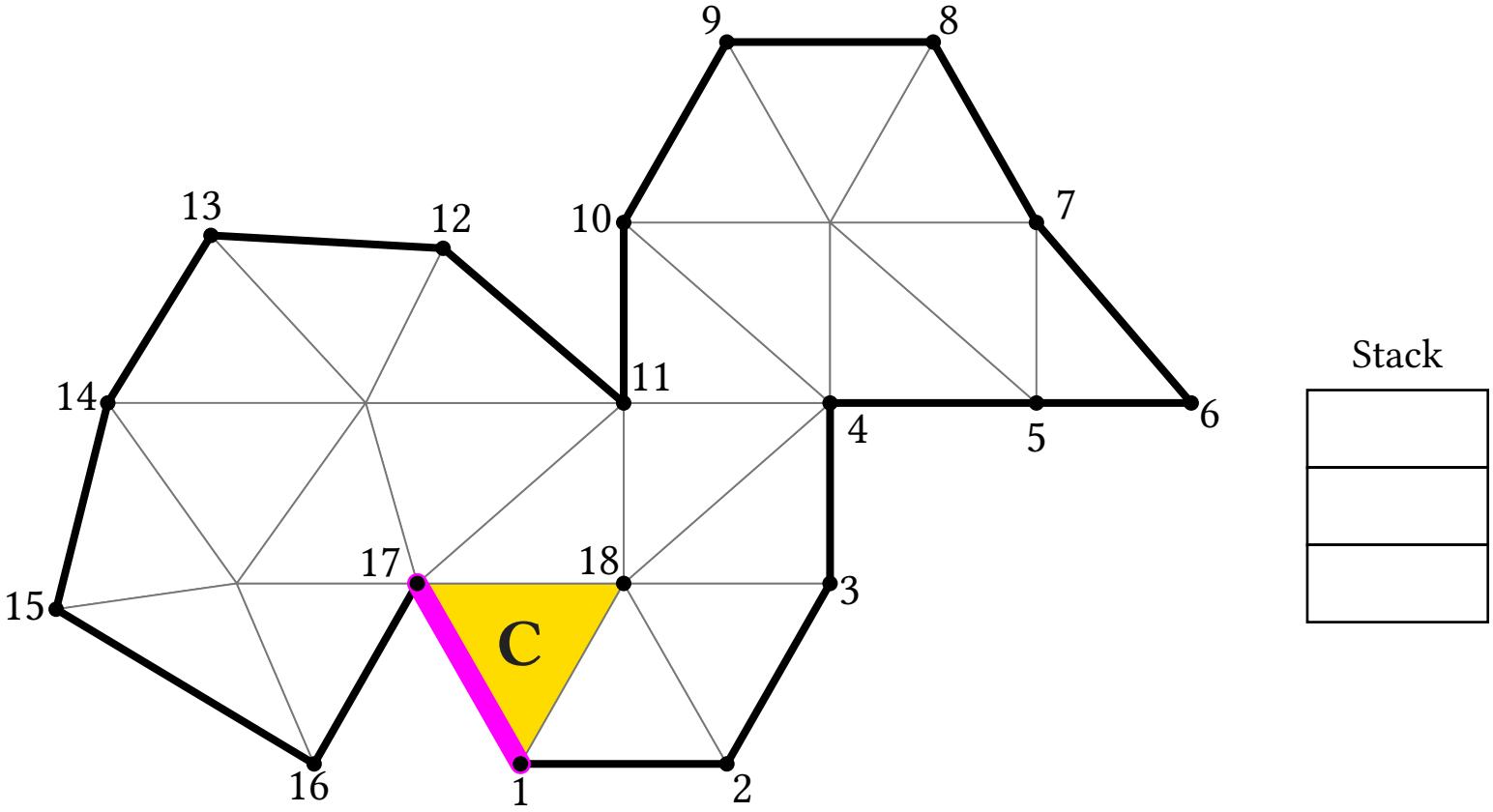


Figure 4: Jednoduchá komprese.

History = CCR

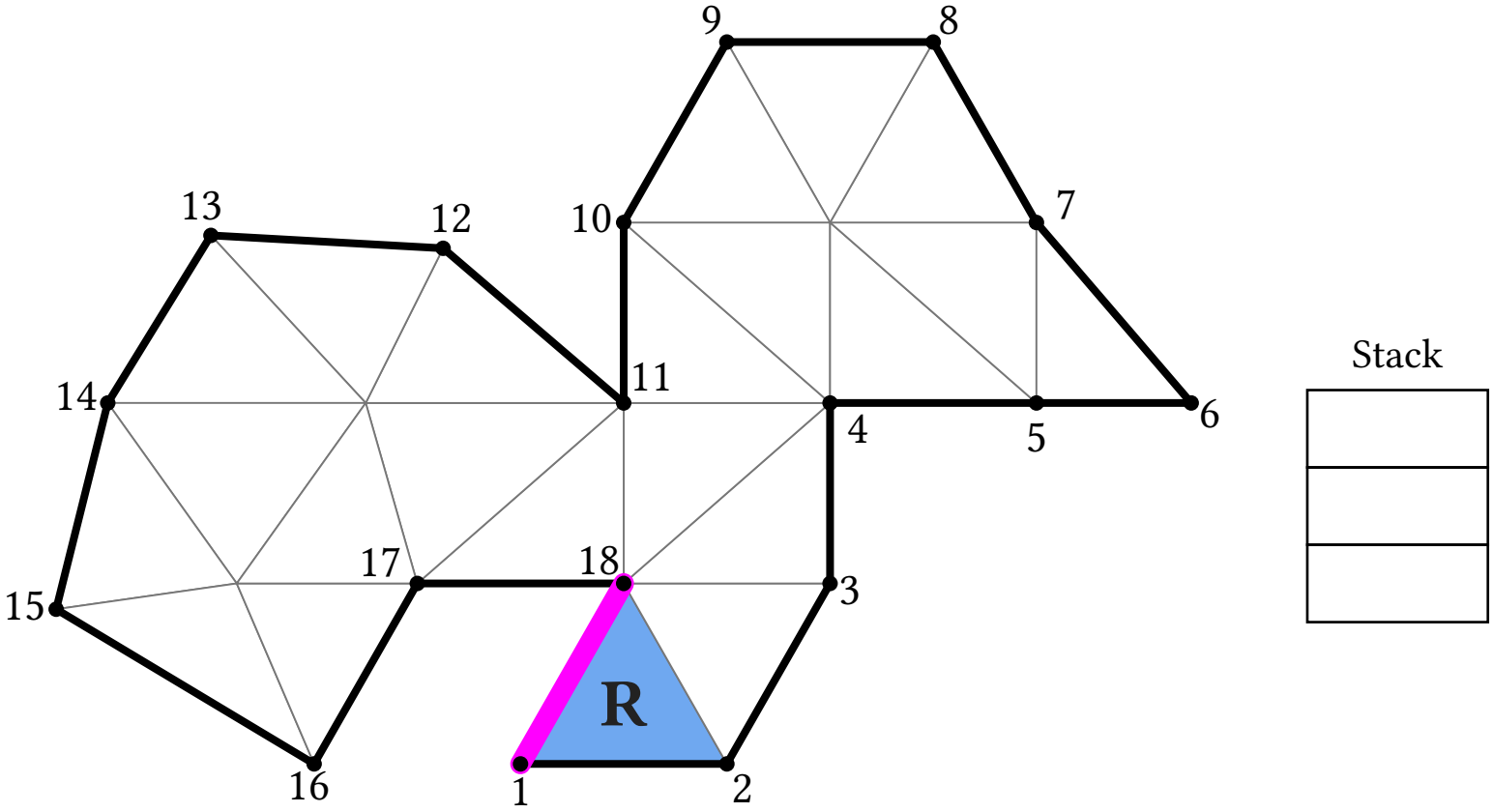


Figure 4: Jednoduchá komprese.

History = CCRR

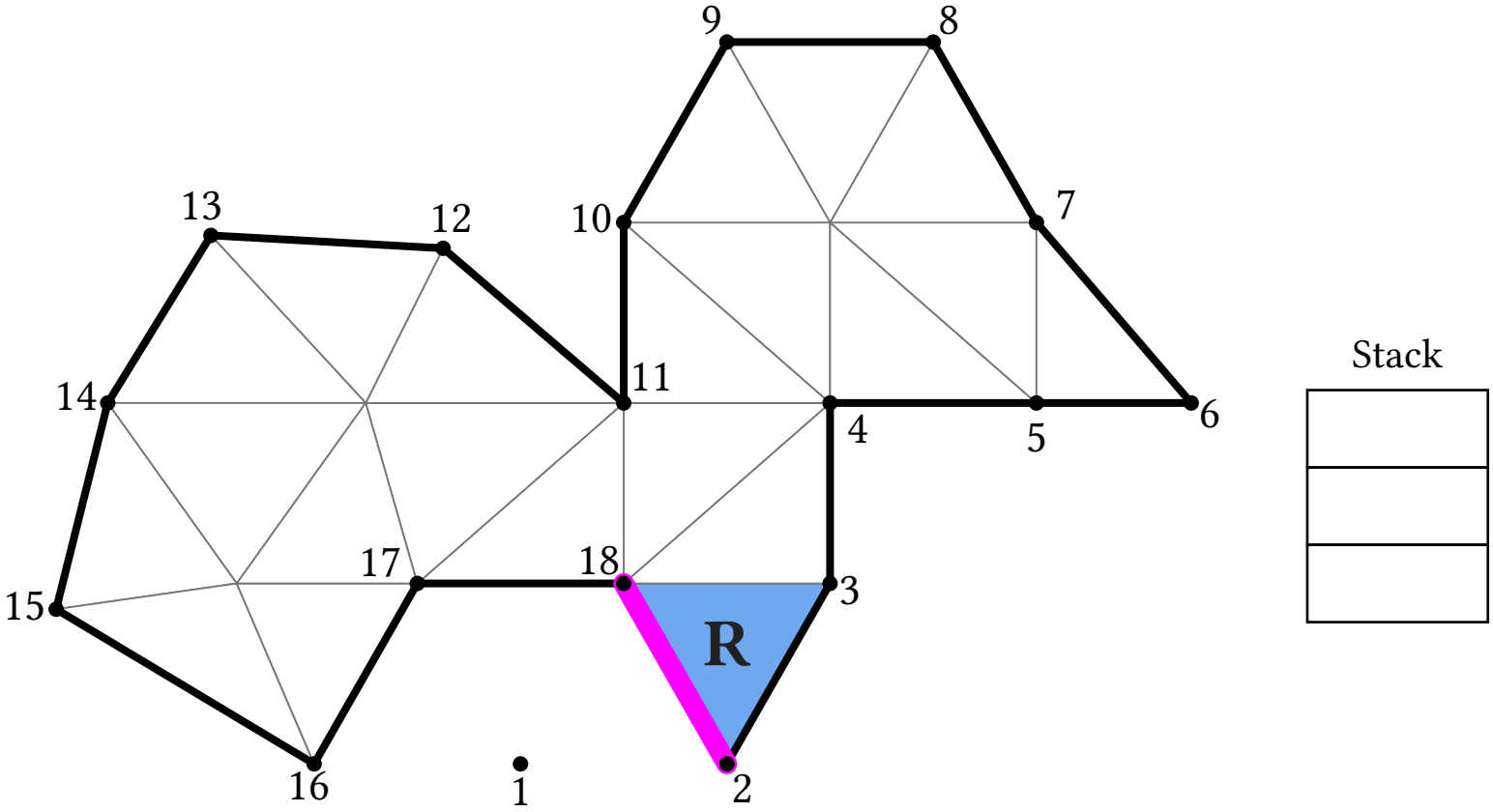


Figure 4: Jednoduchá komprese.

History = CCRRR

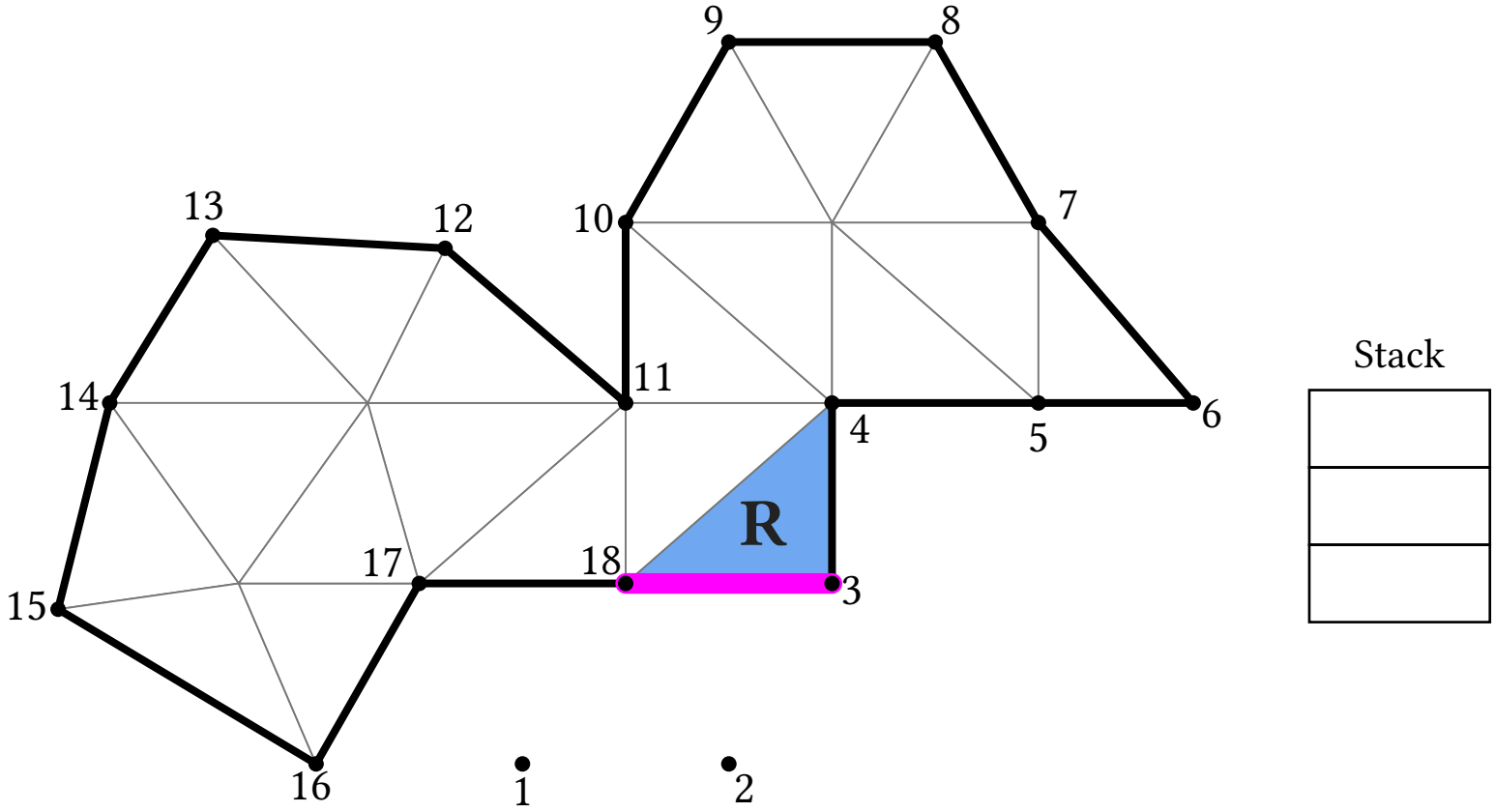


Figure 4: Jednoduchá komprese.

History = CCRRRS

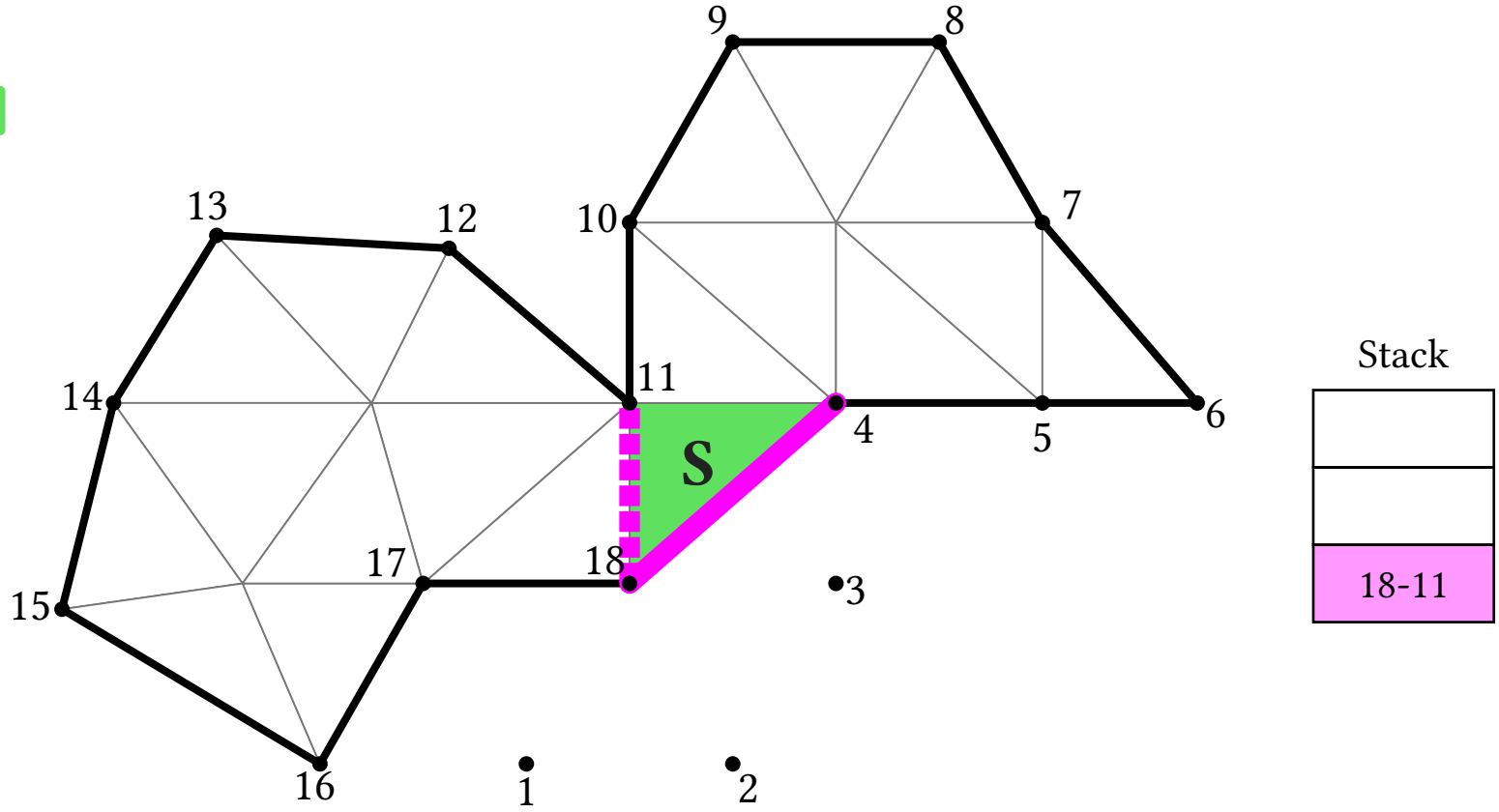


Figure 4: Jednoduchá komprese.

History = CCRRRSL

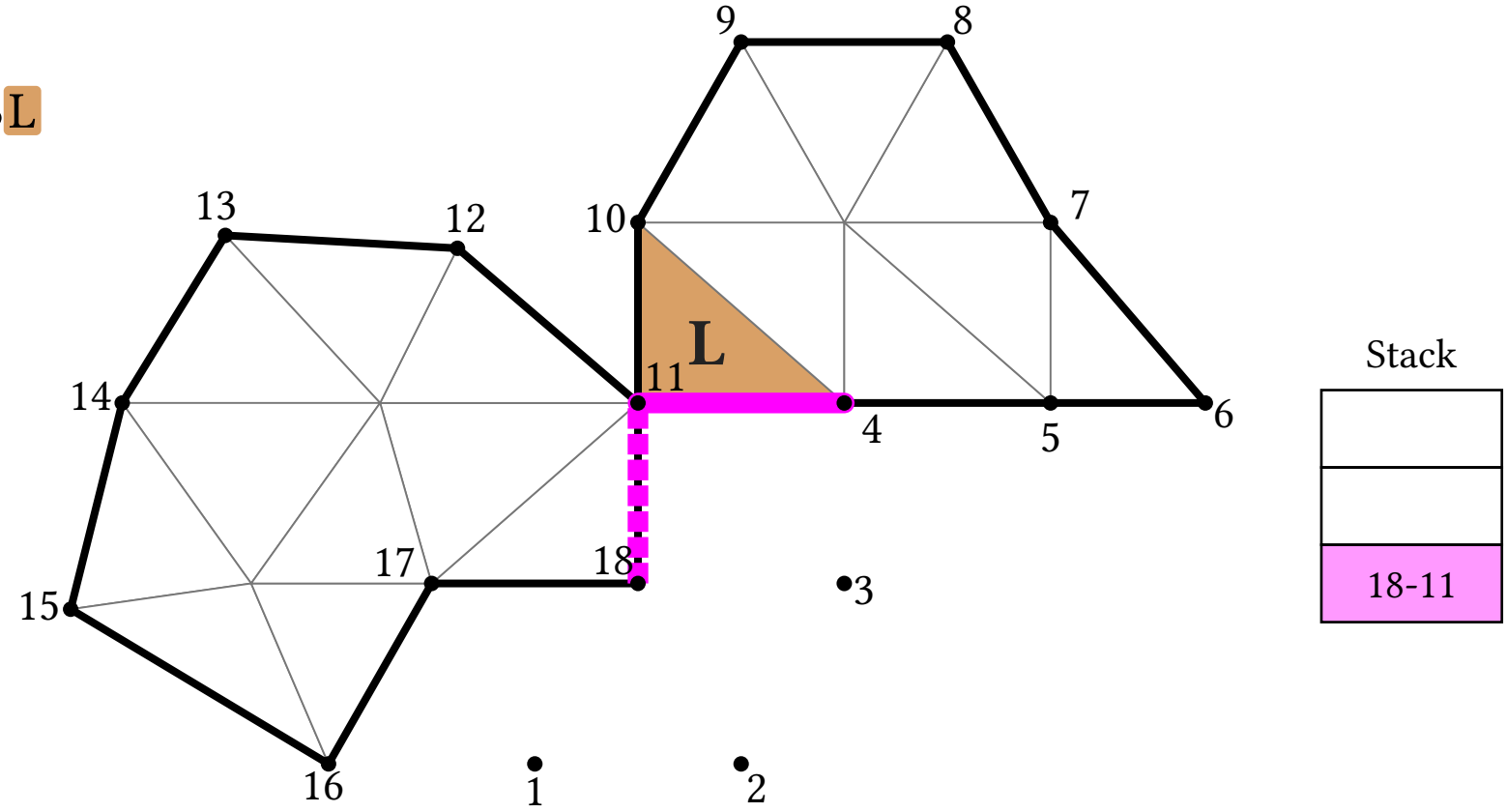


Figure 4: Jednoduchá komprese.

History = CCRRRSLC

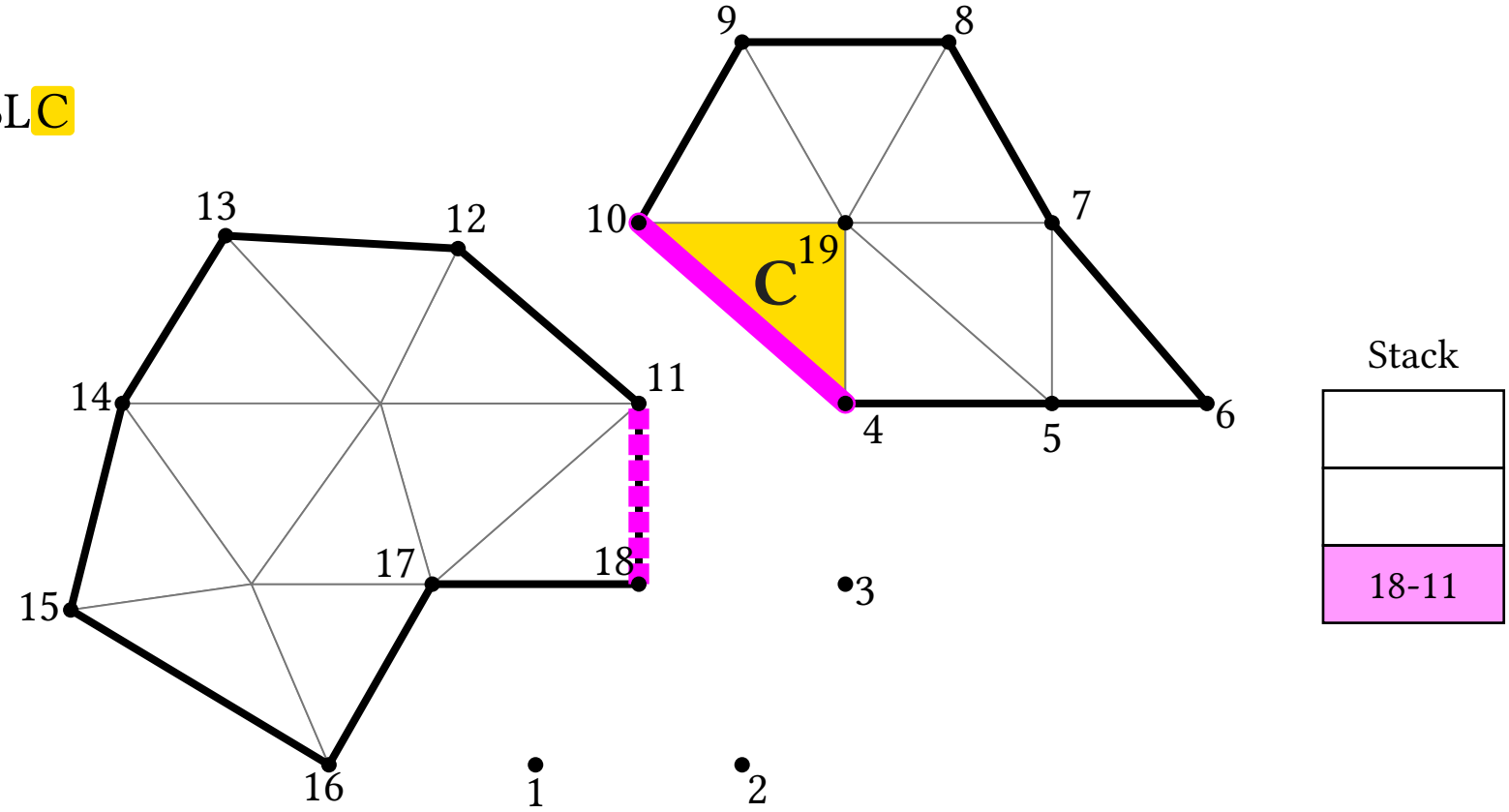


Figure 4: Jednoduchá komprese.

History = CCRRRSLCR

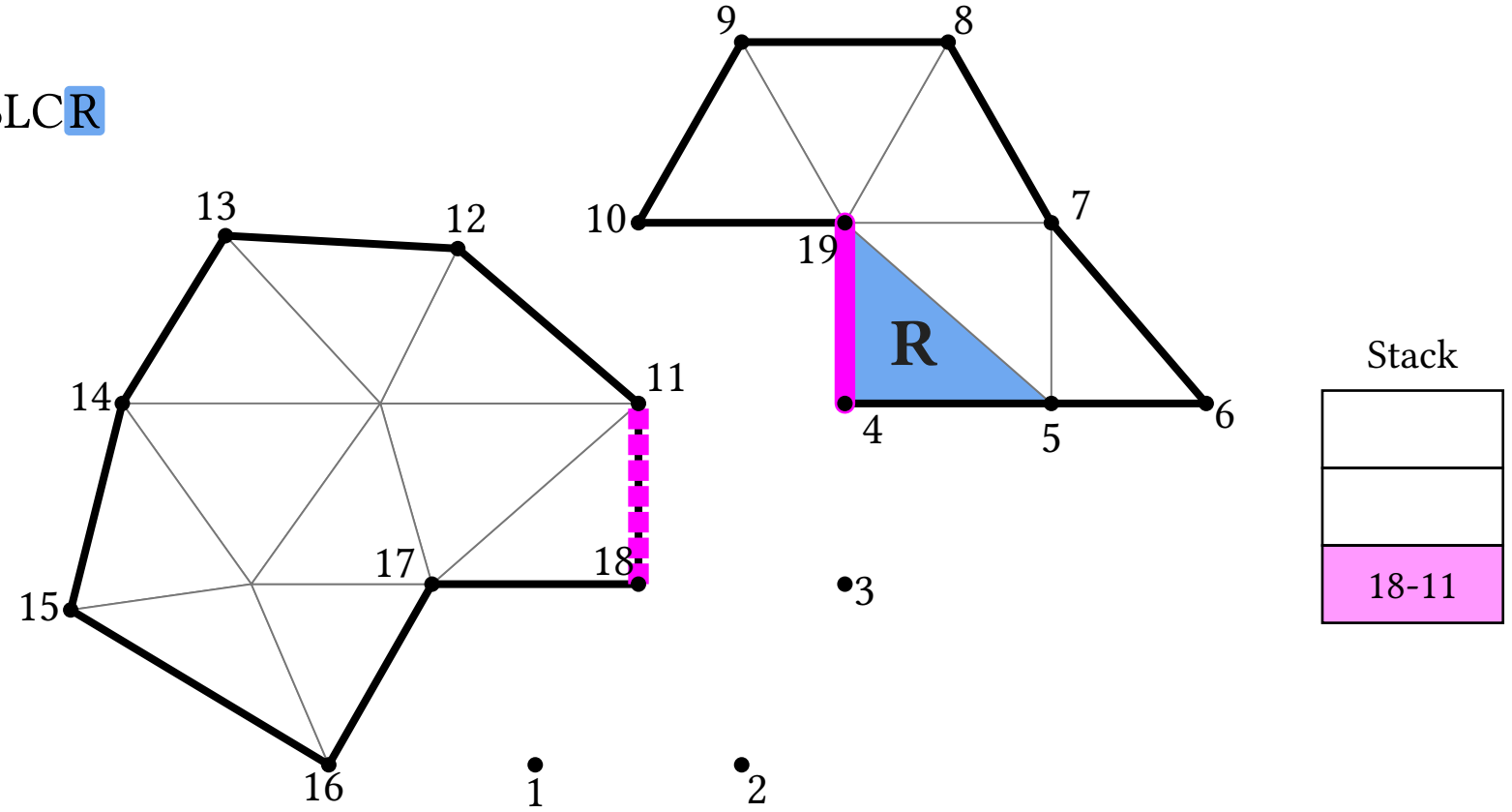


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRS

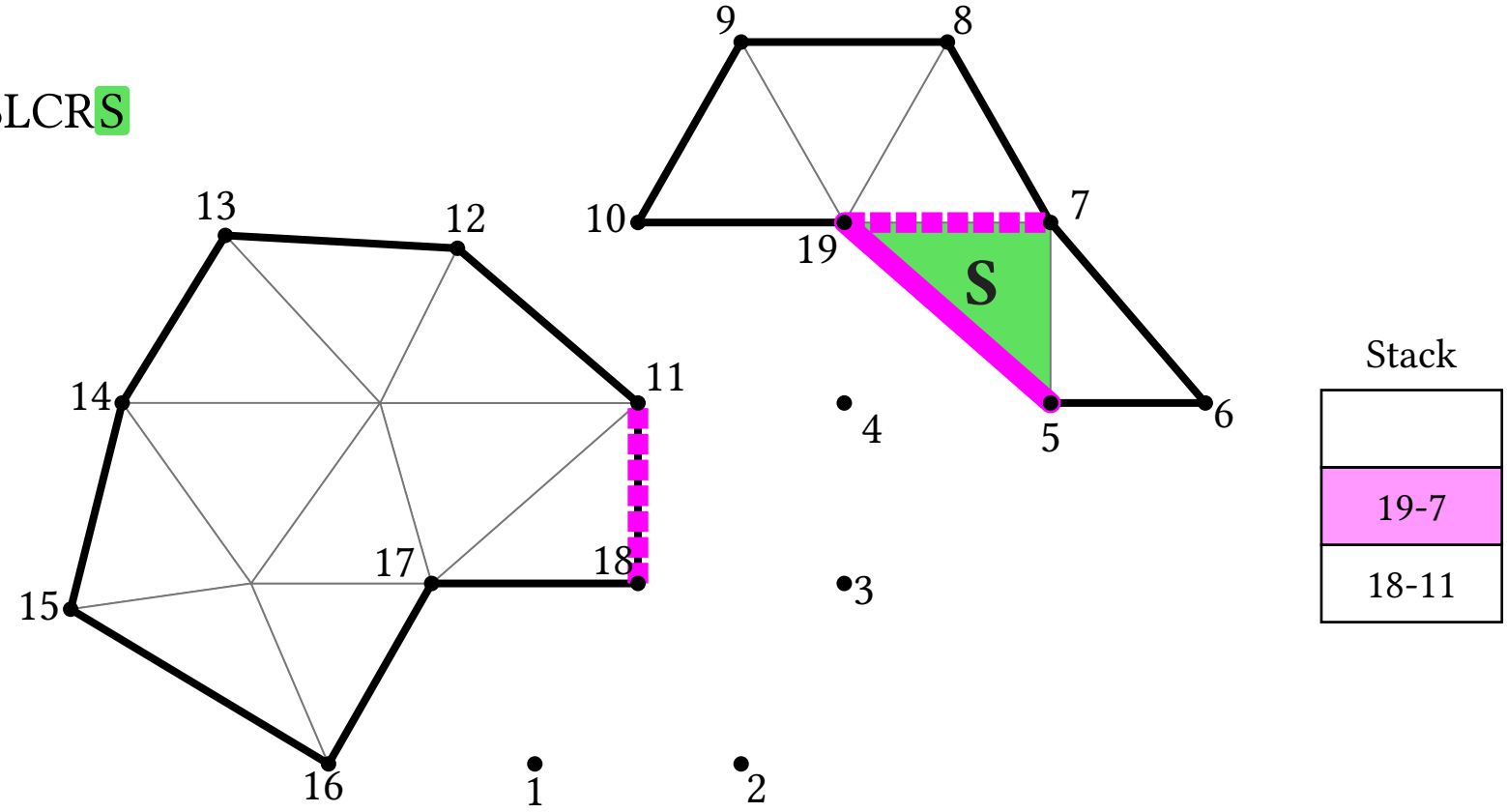


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSE

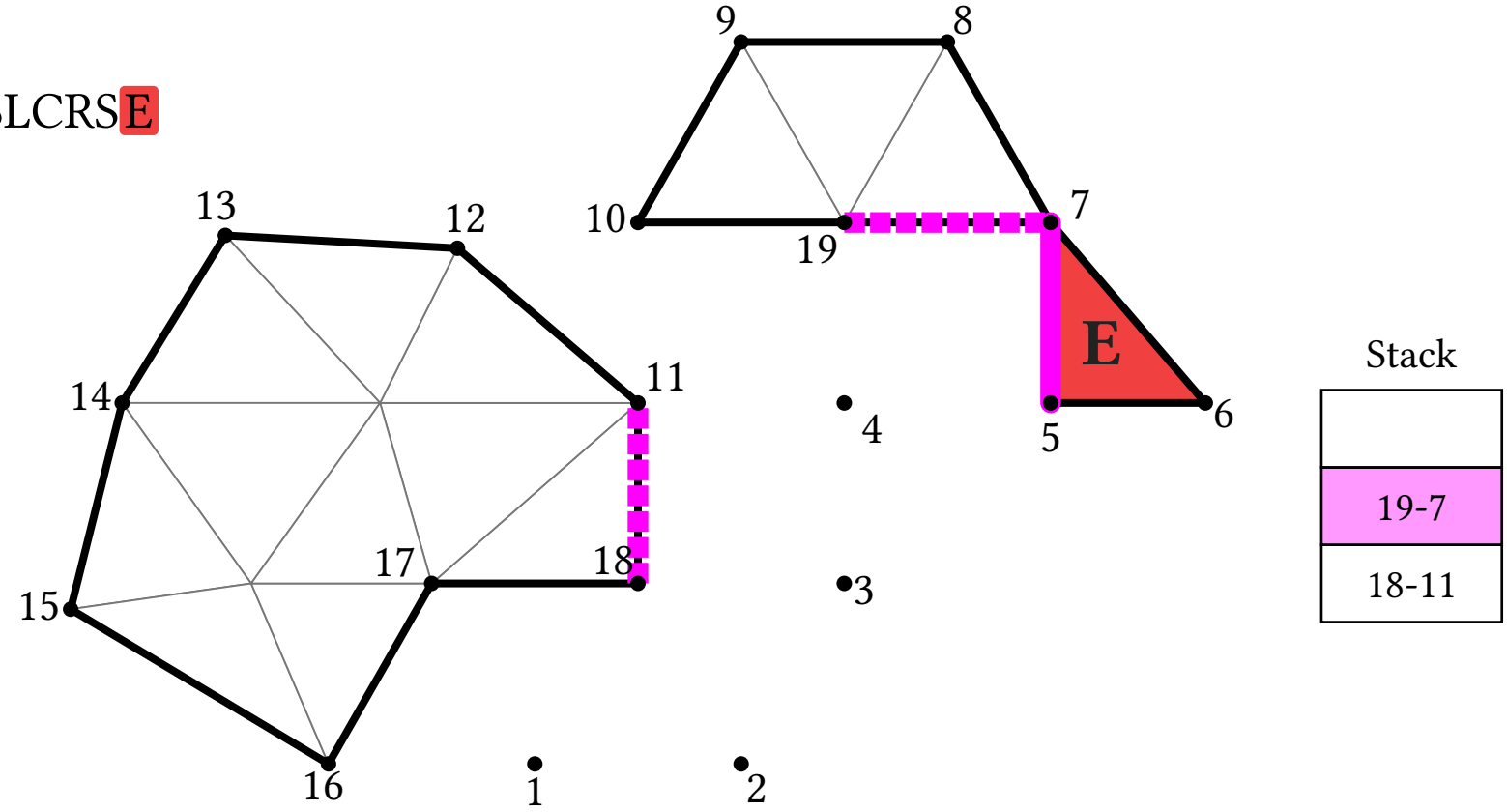


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSER

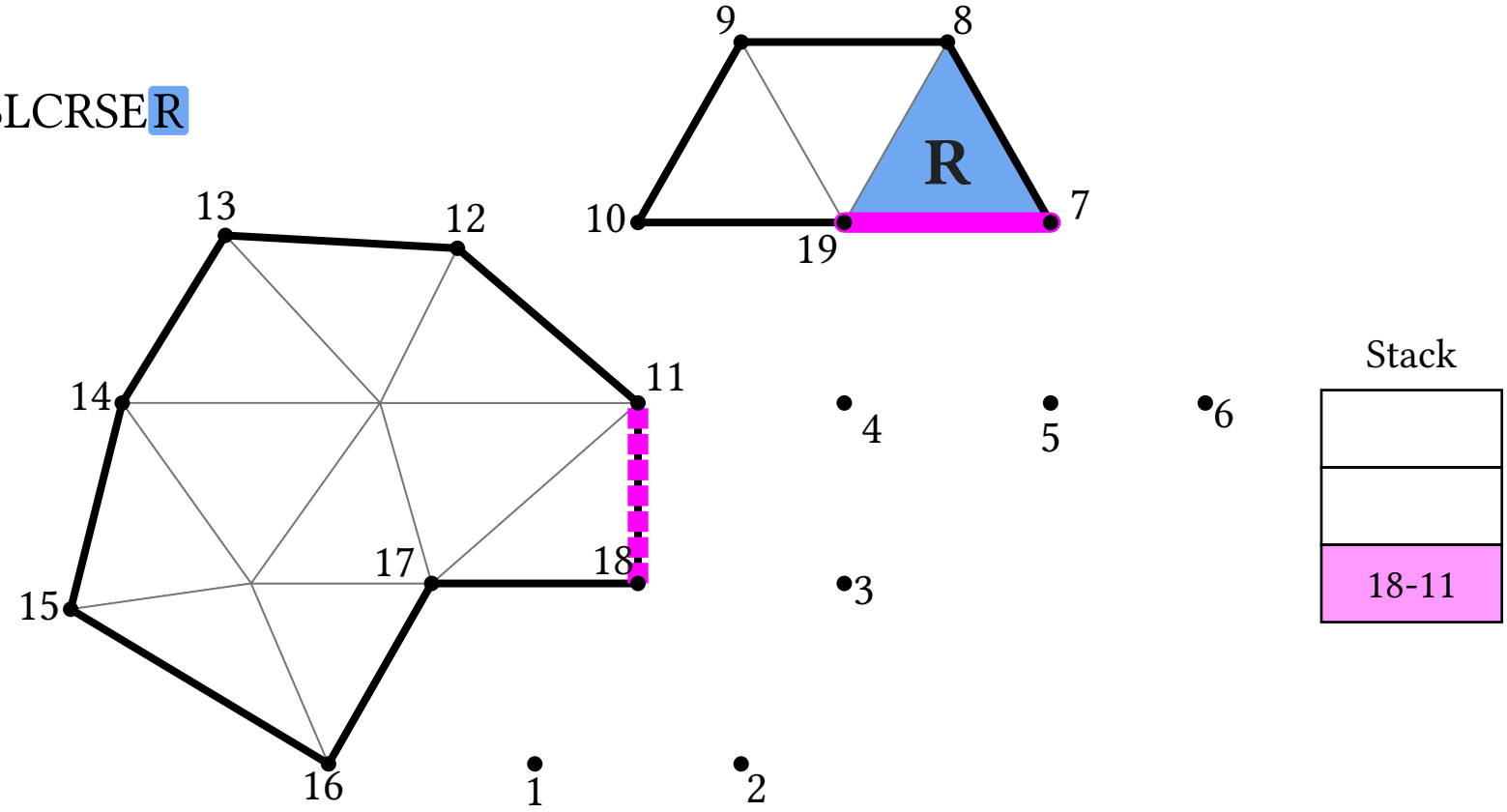


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERR

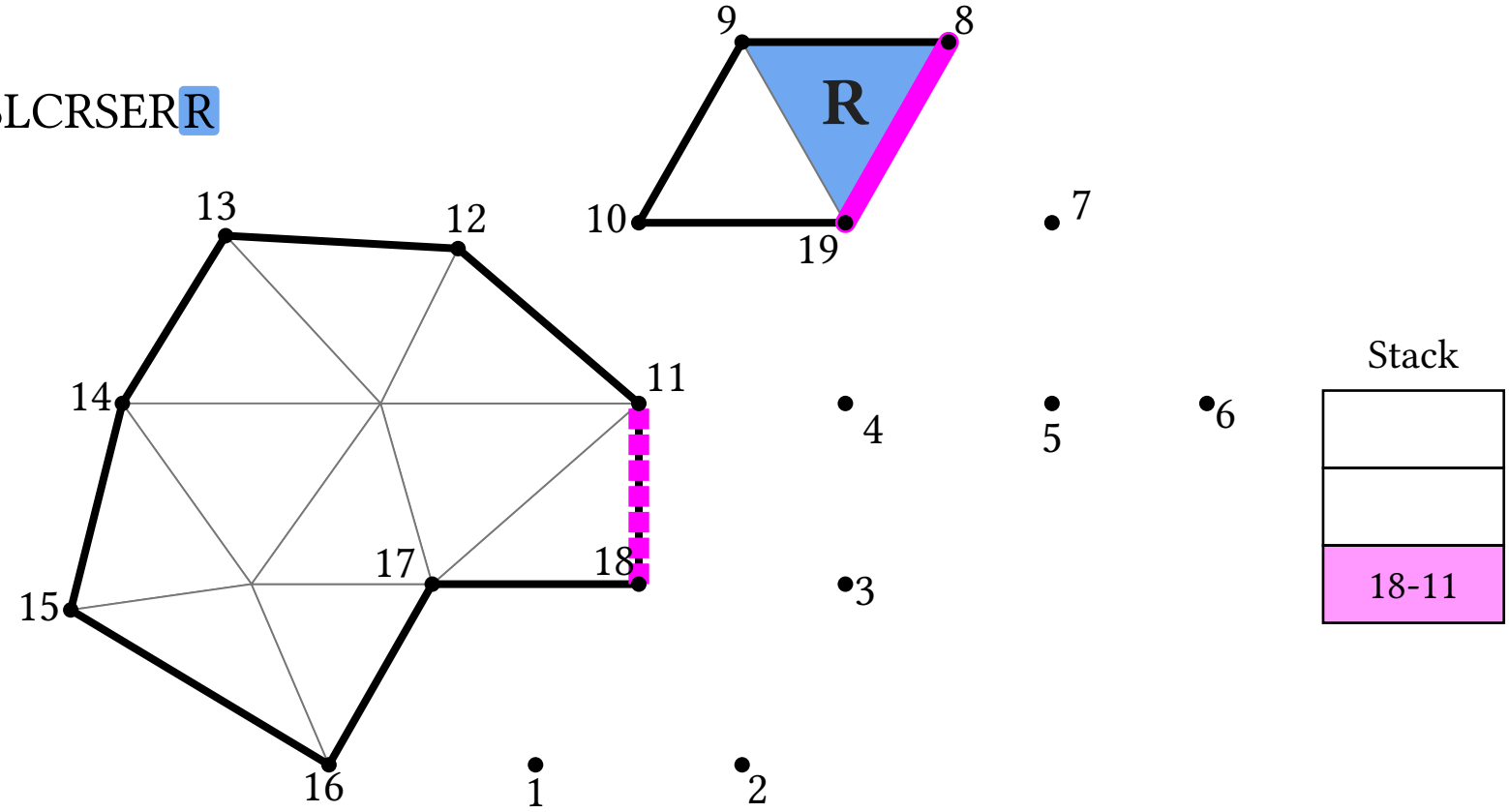


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRE

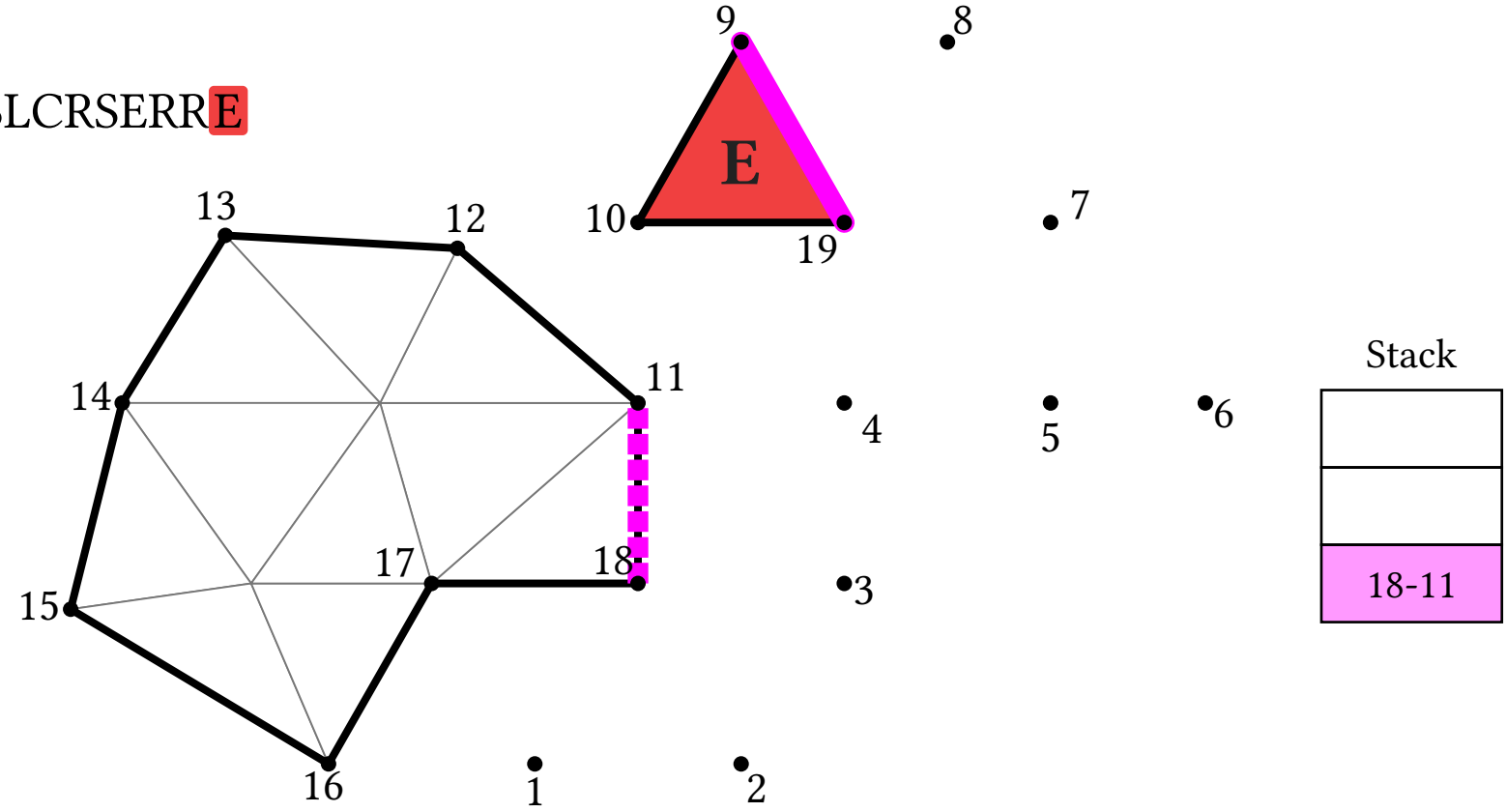


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERREL

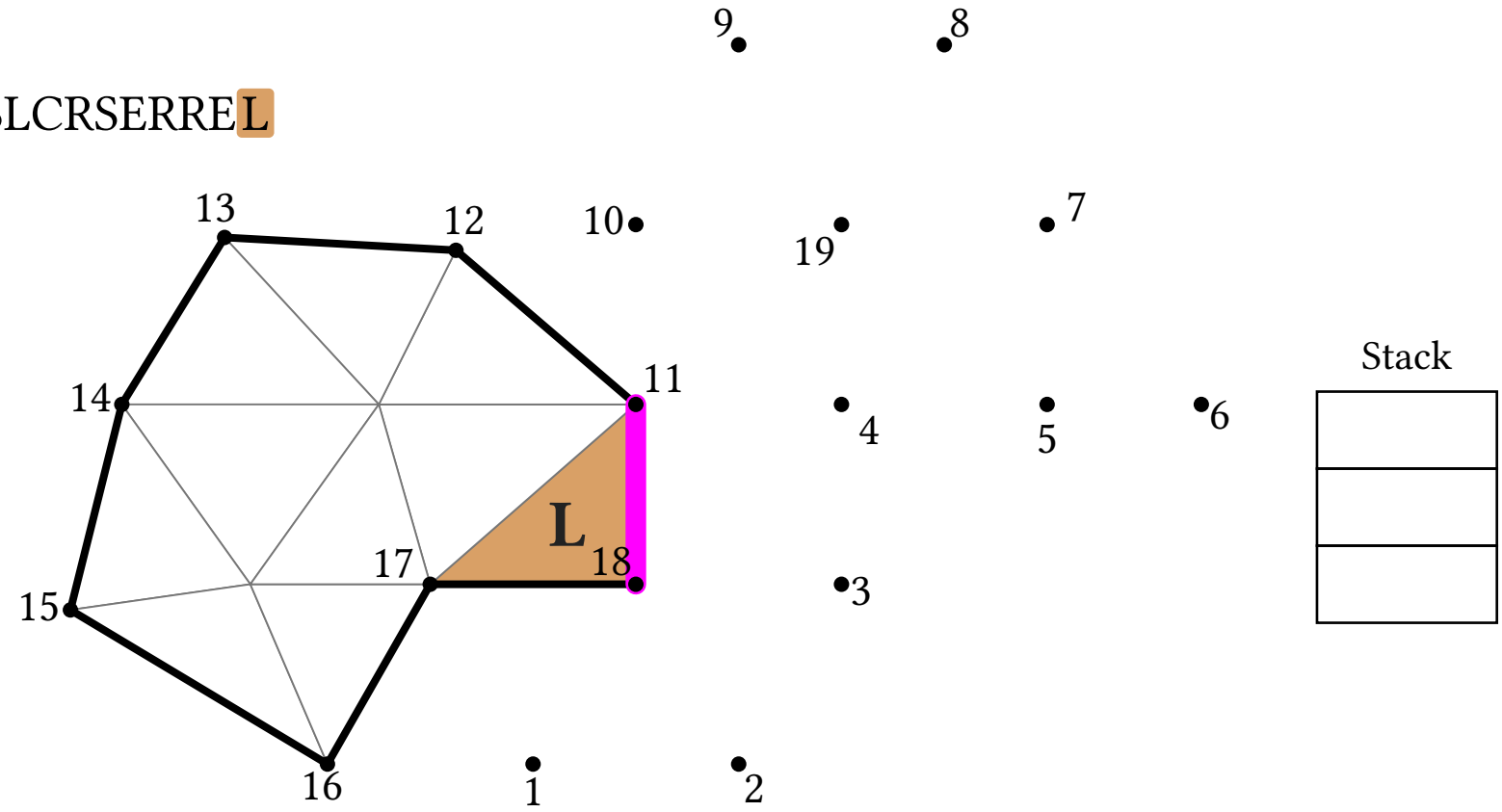


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

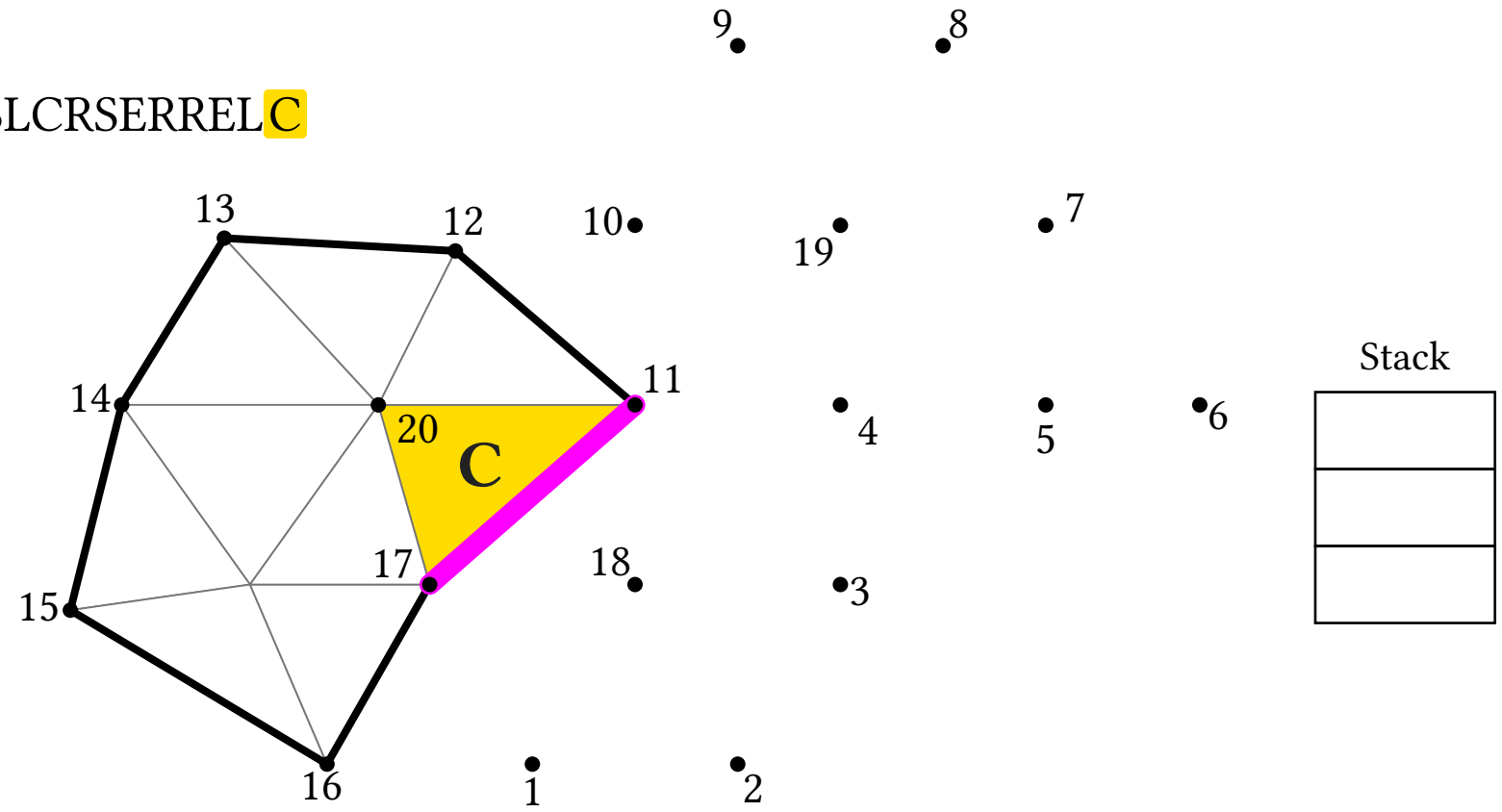


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

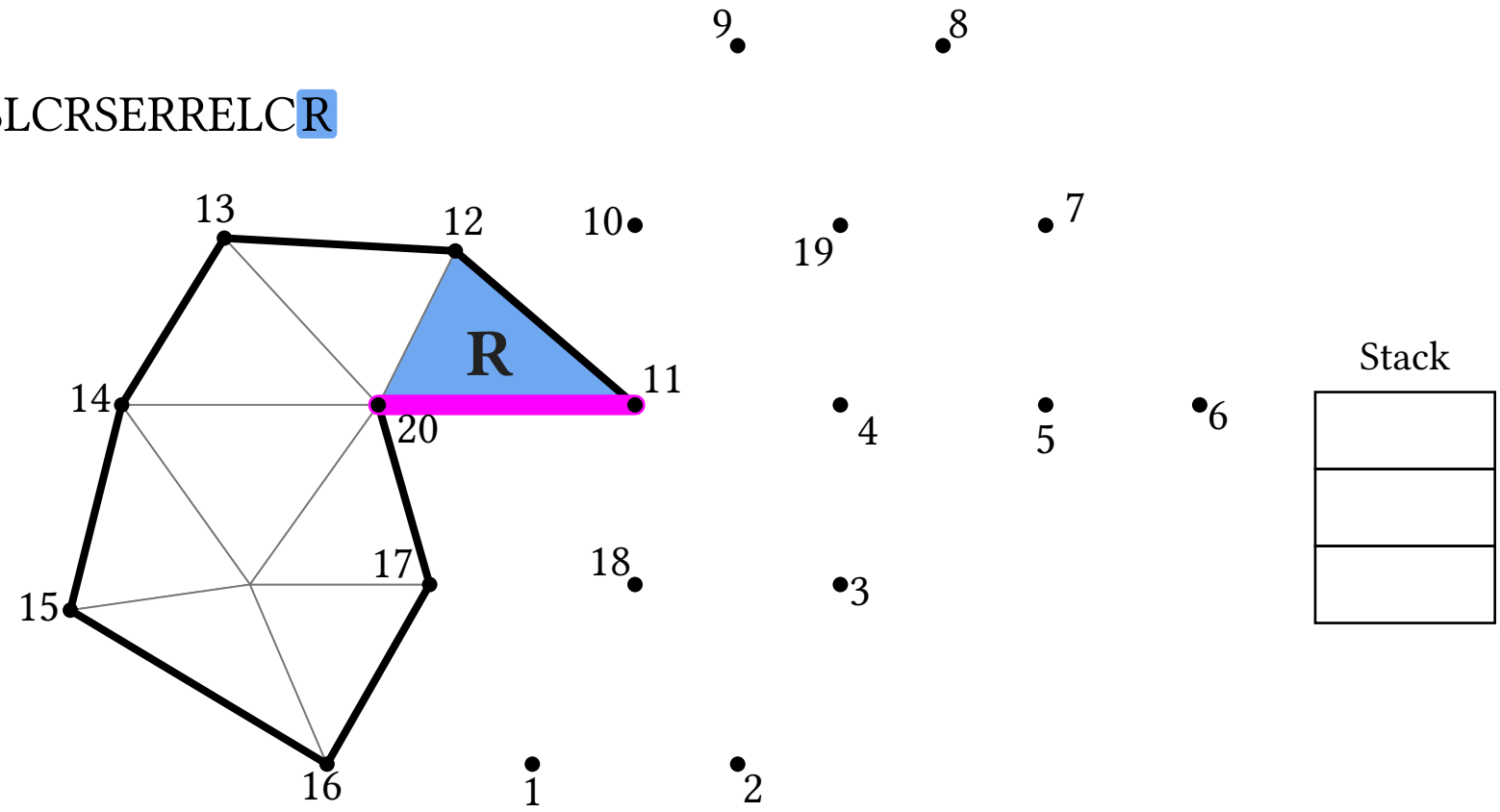


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

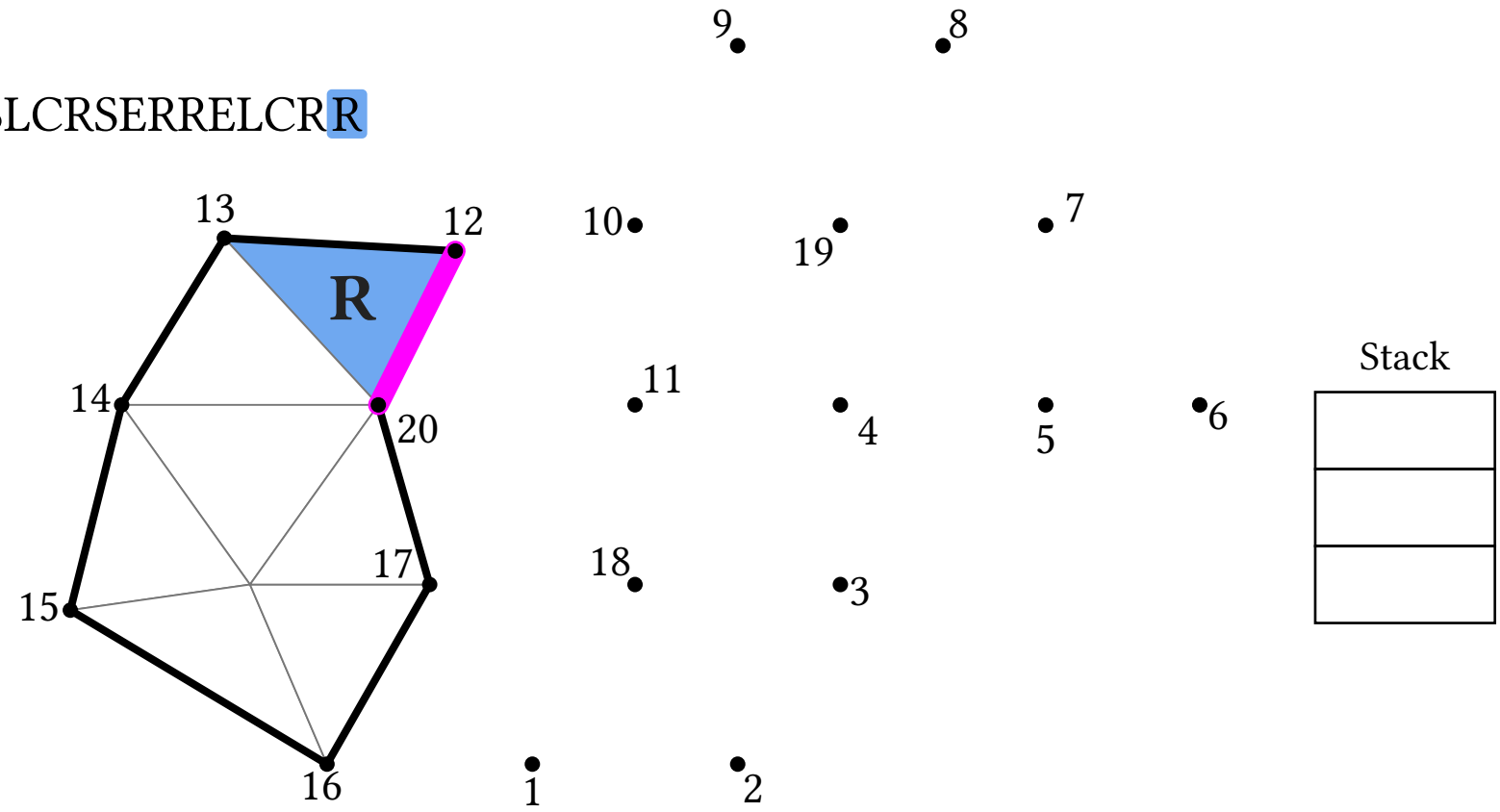


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

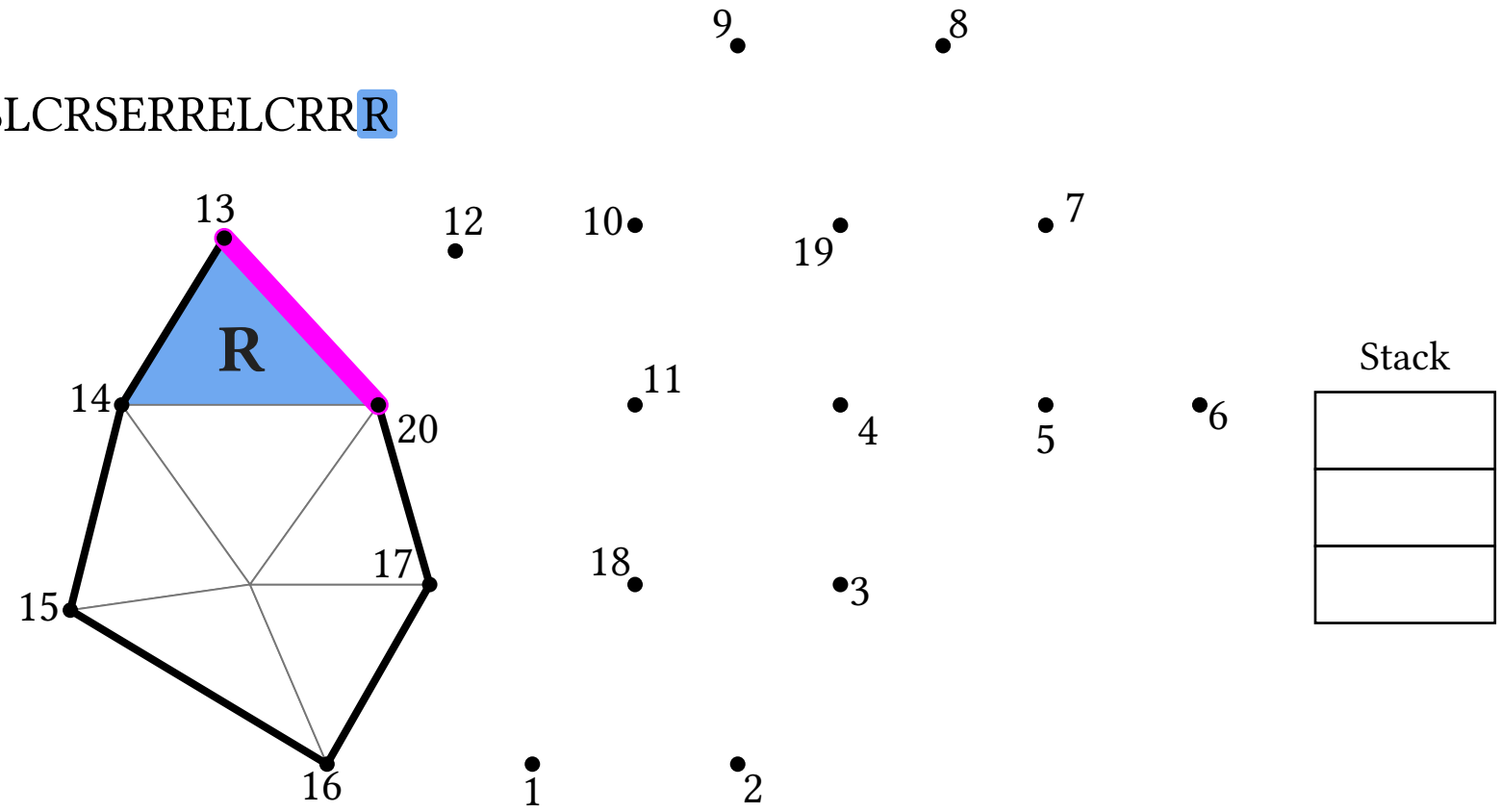


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

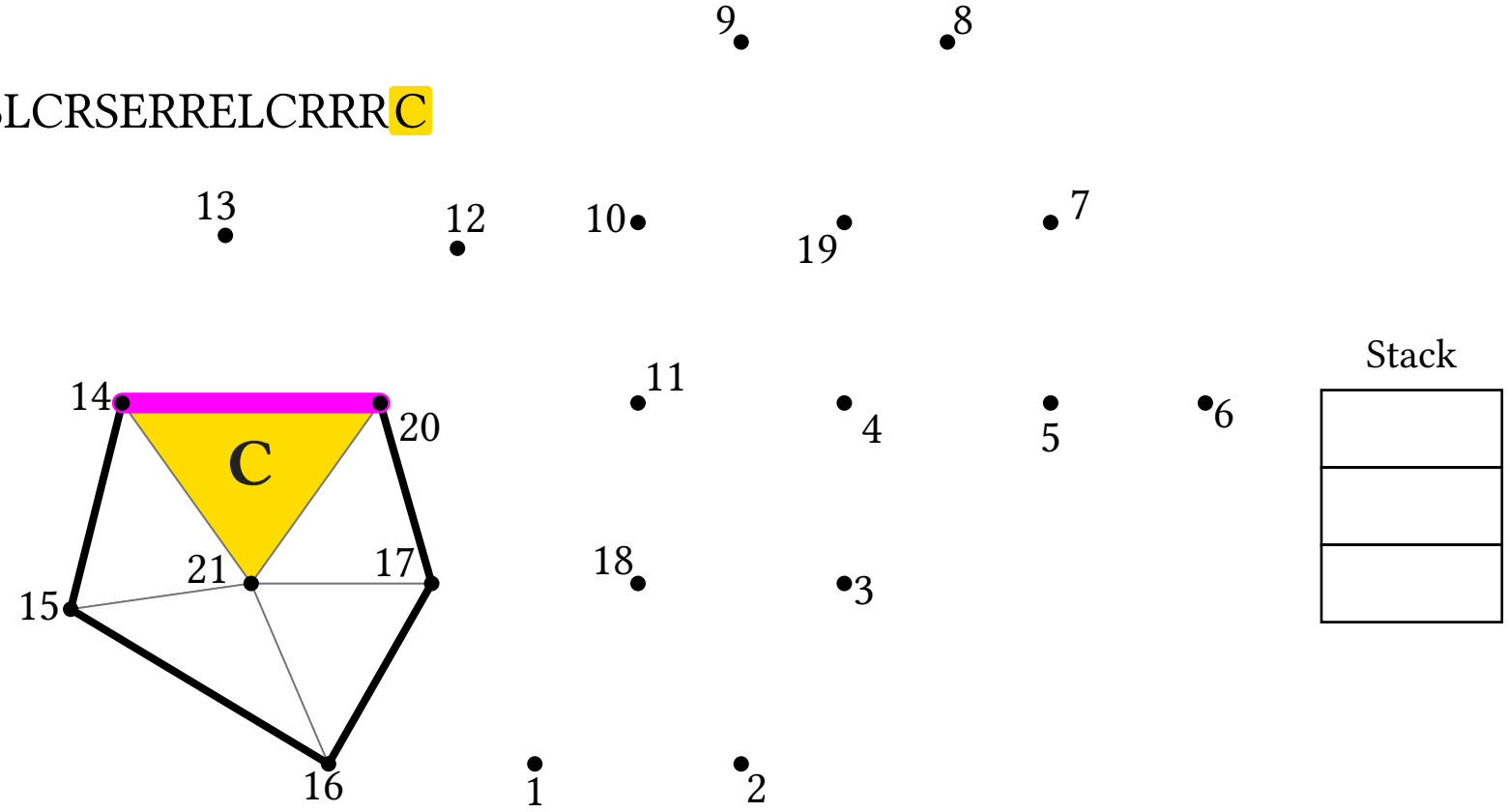


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

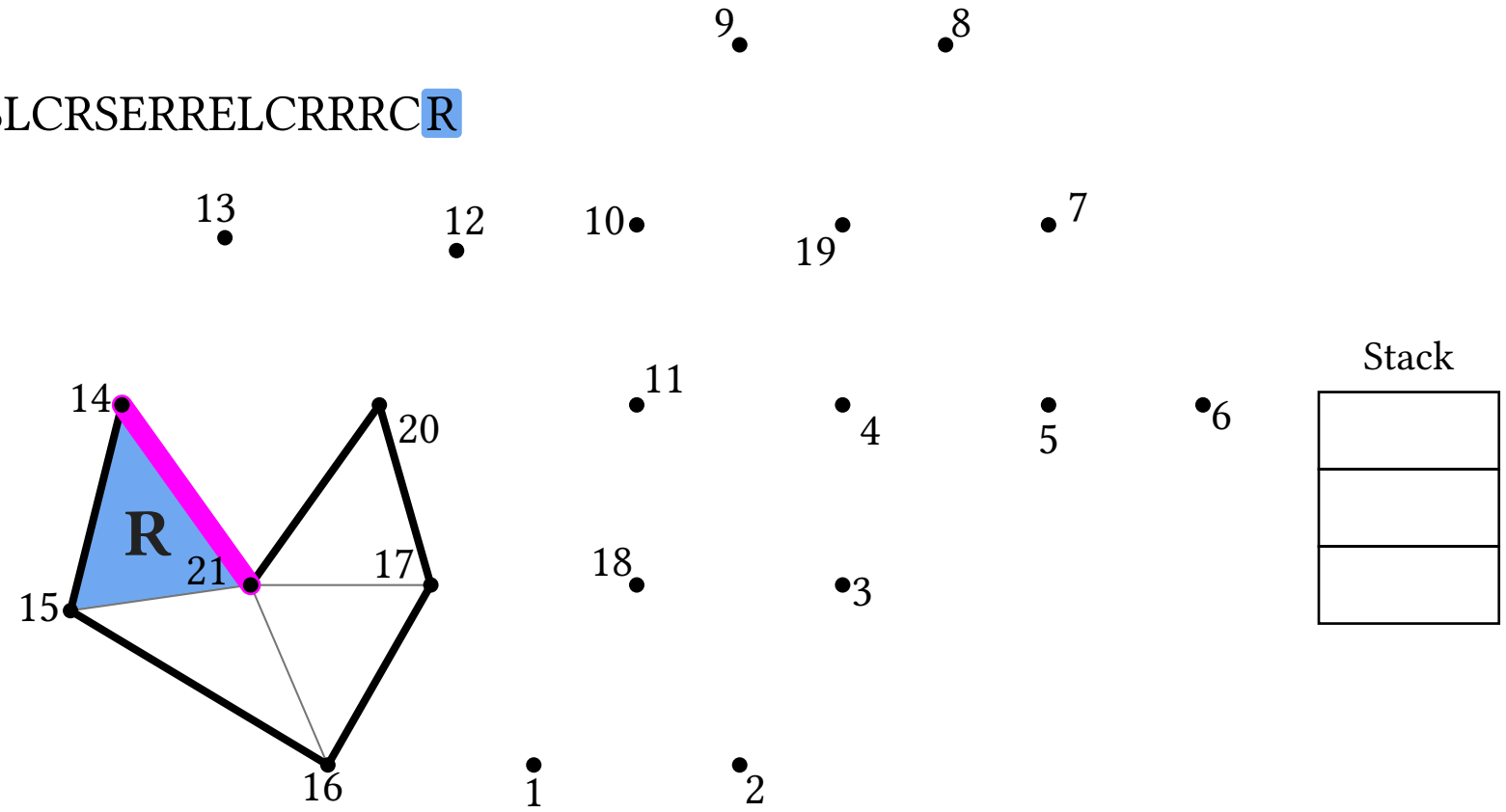


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

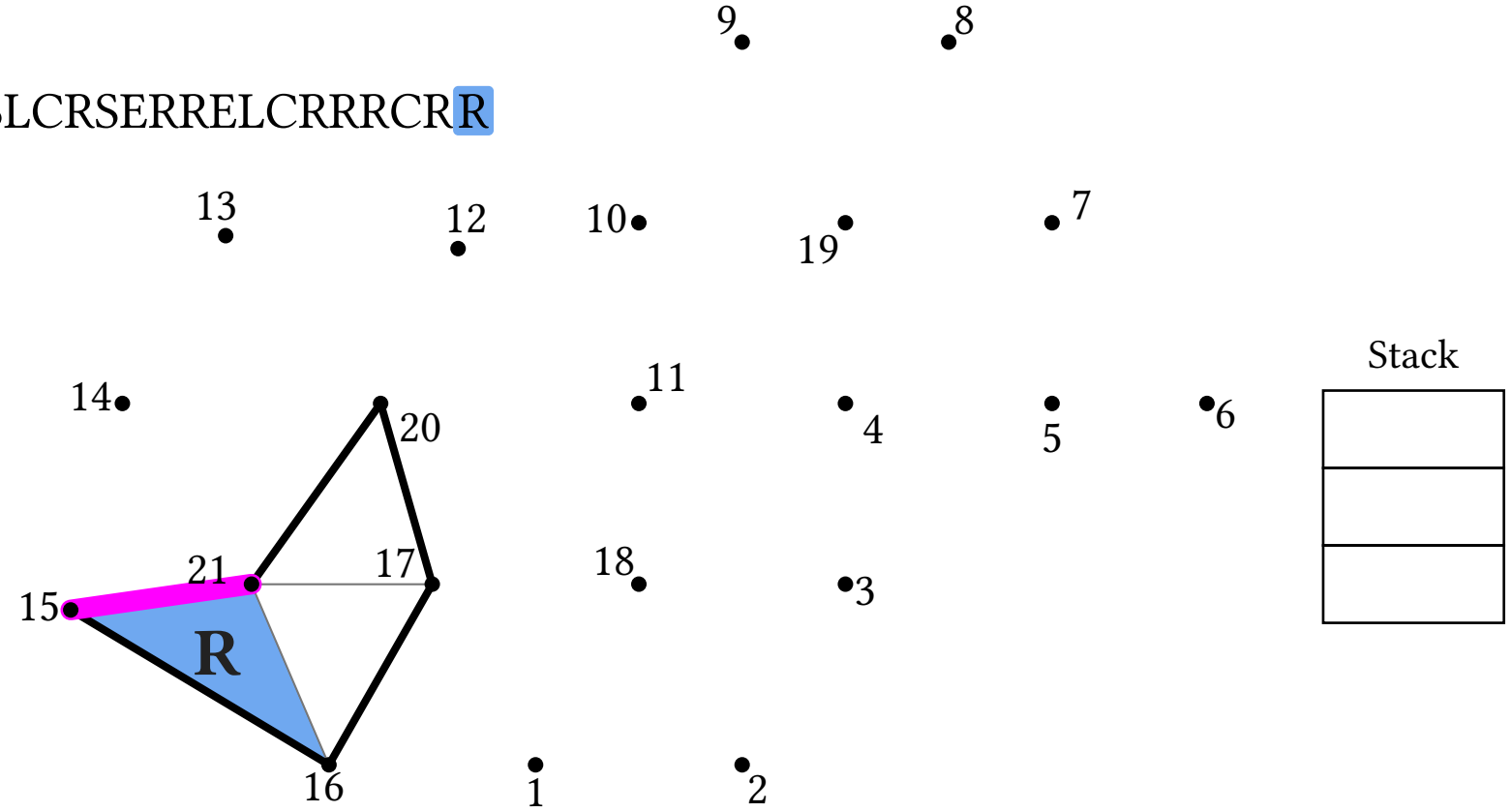


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

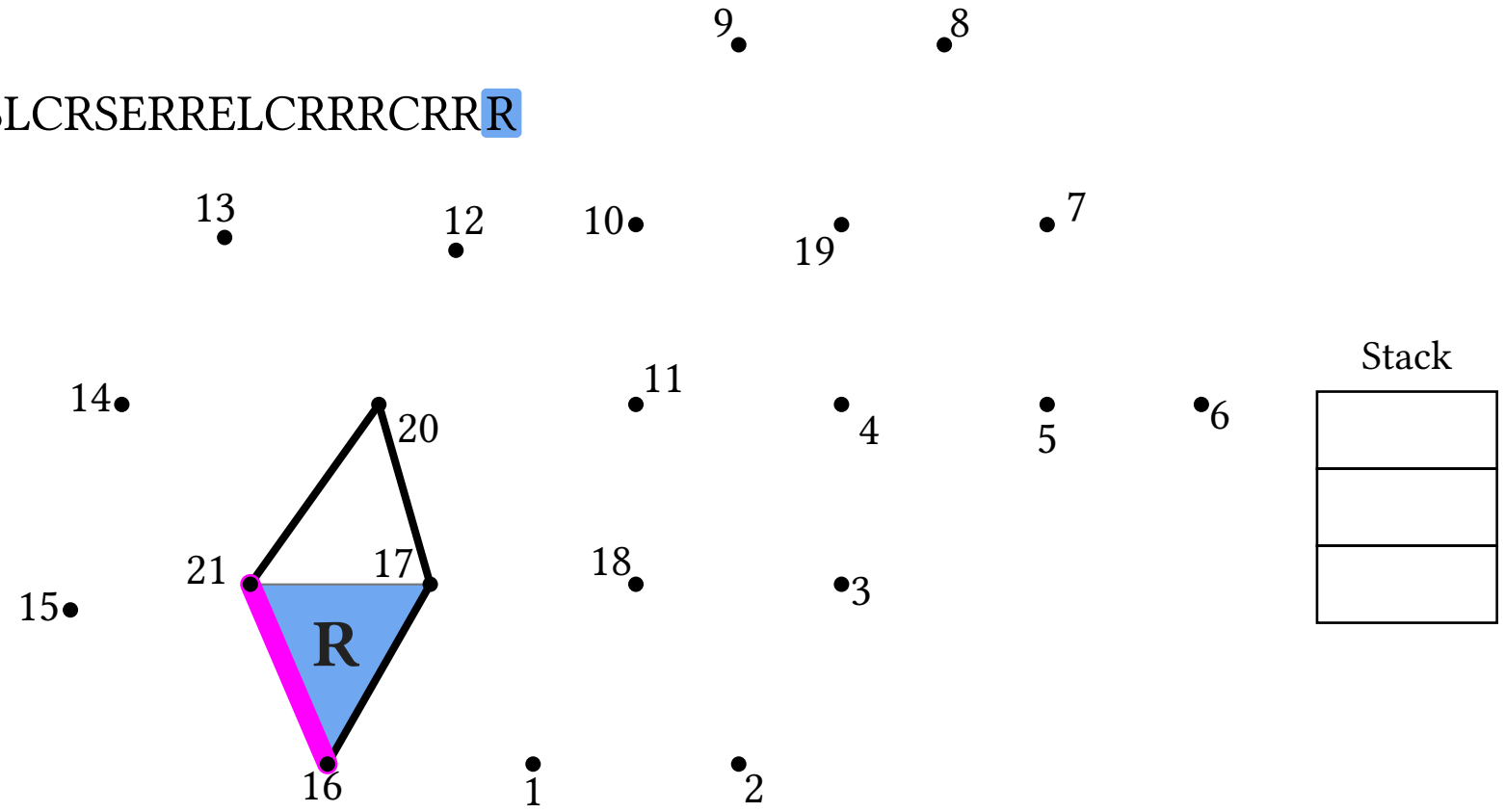


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

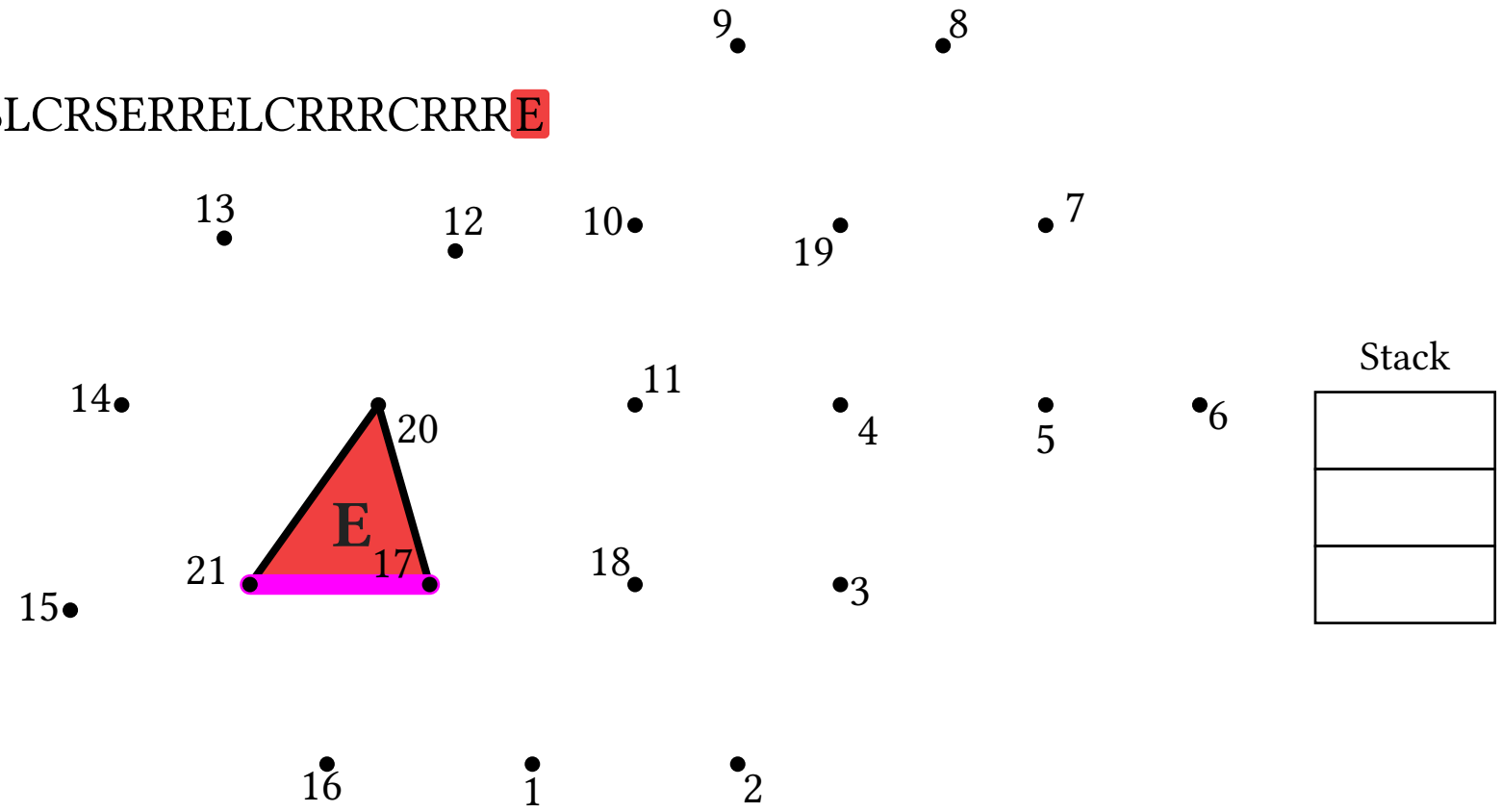


Figure 4: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

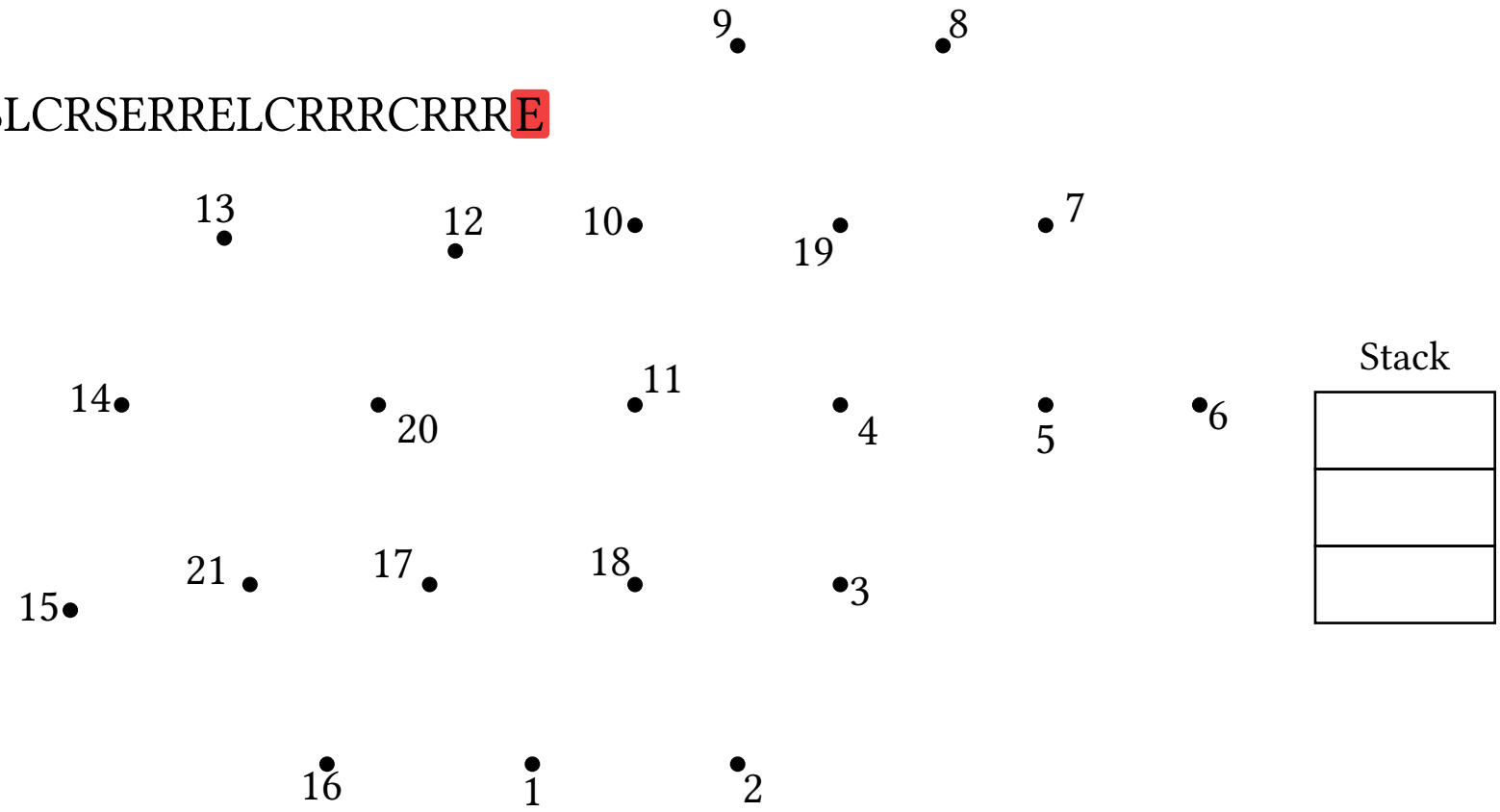


Figure 4: Jednoduchá komprese.

3.3 Vyšší genus?

History =

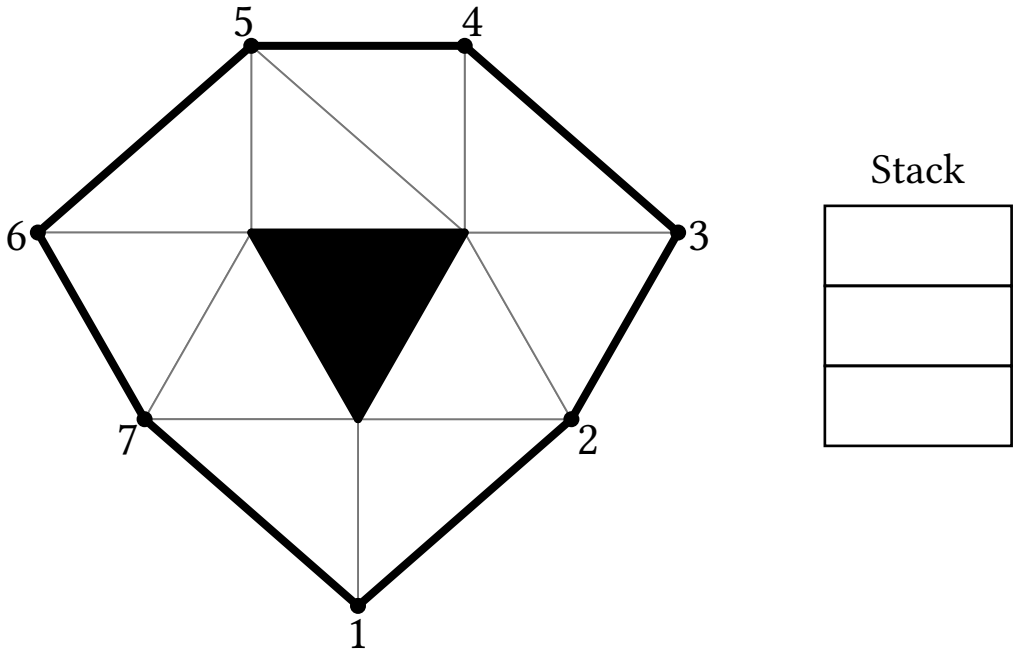


Figure 5: Komprese díry.

3.3 Vyšší genus?

History = S

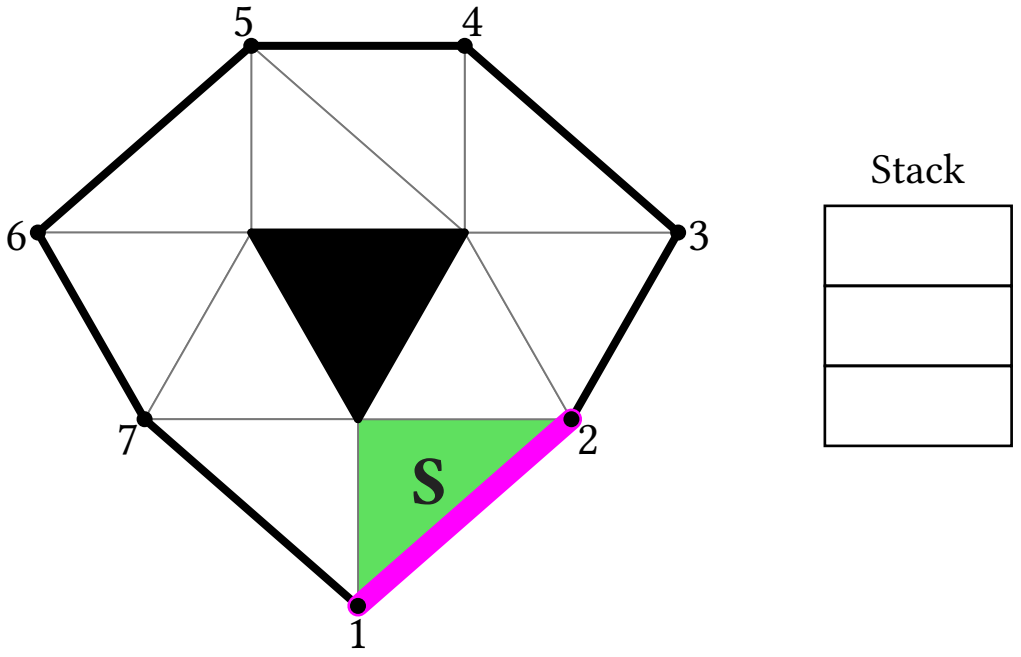


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = S

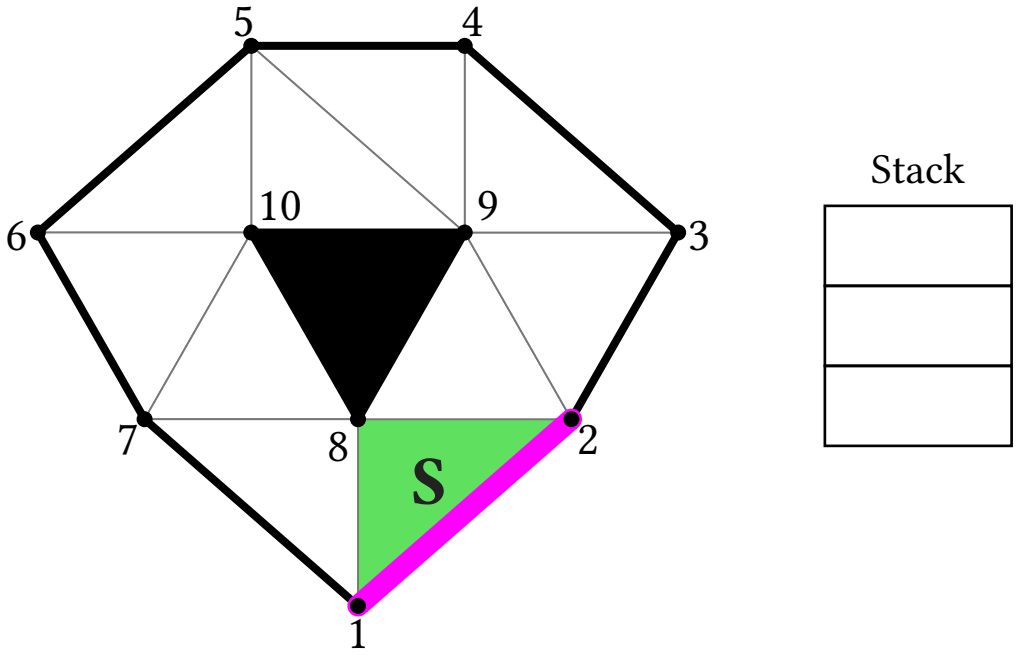


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SL

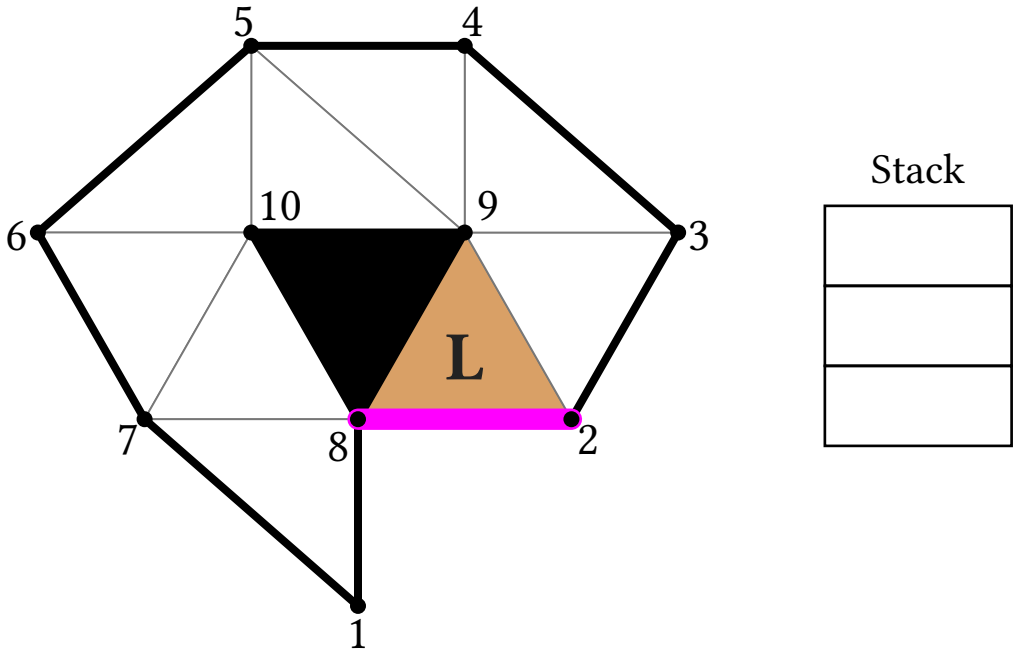


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLR

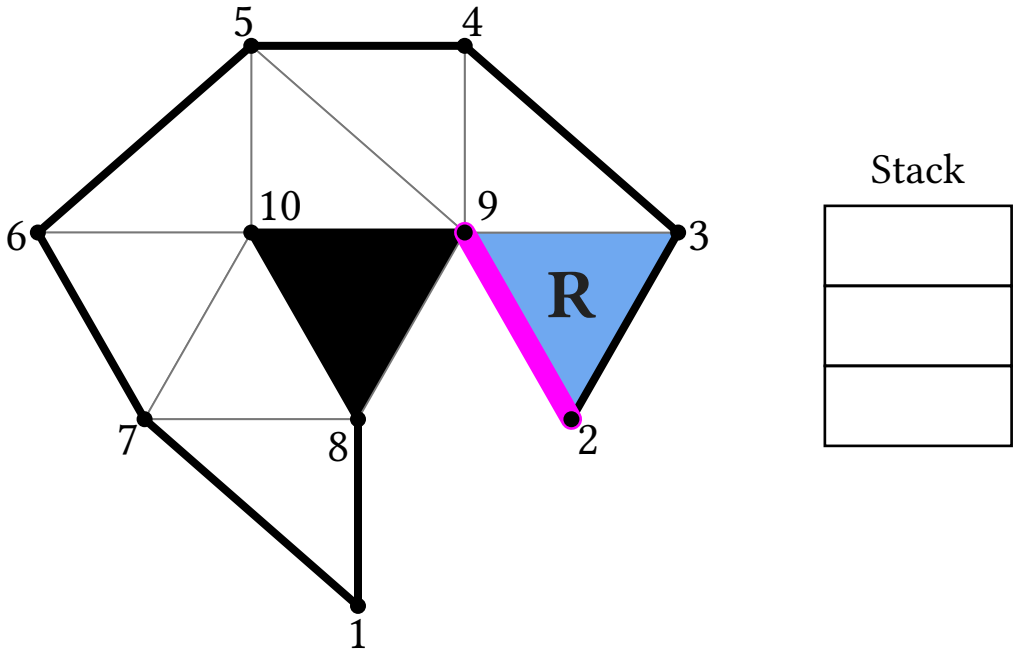


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRR

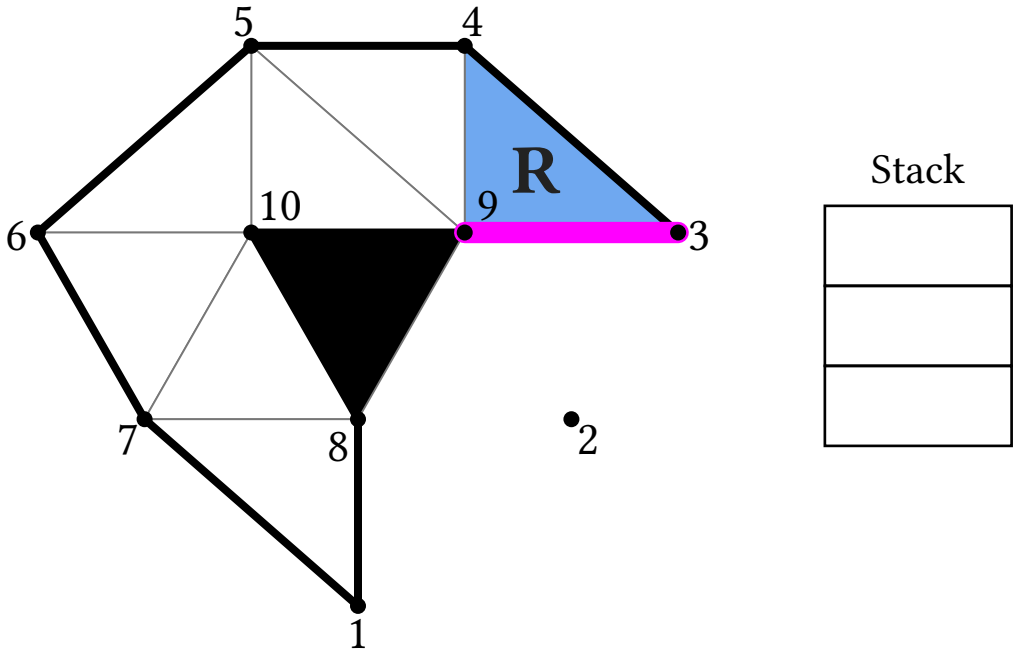


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRR

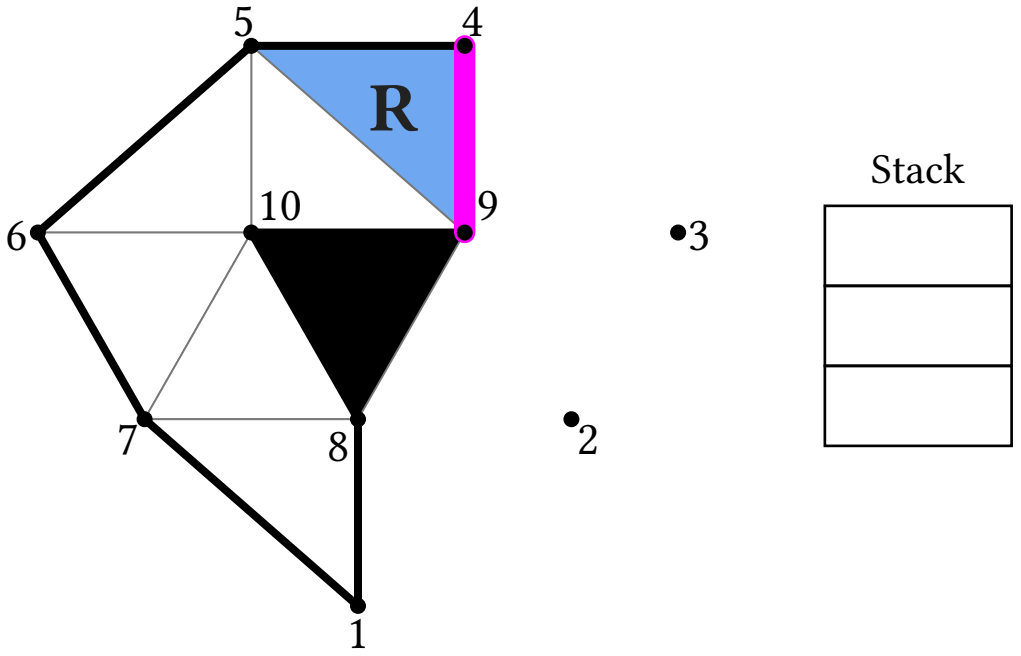


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRL

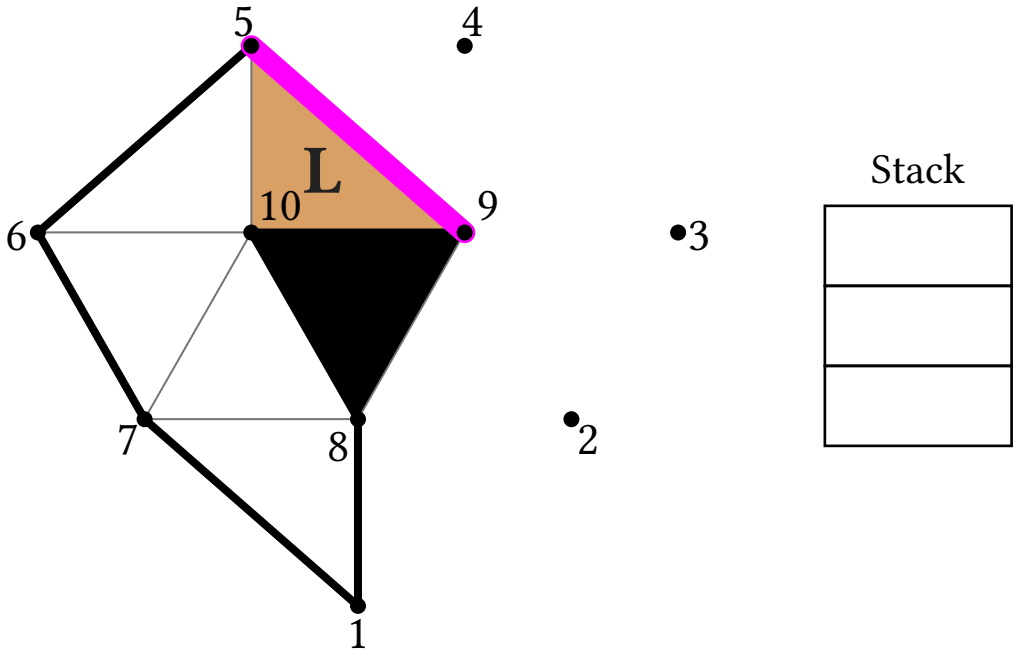


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLR

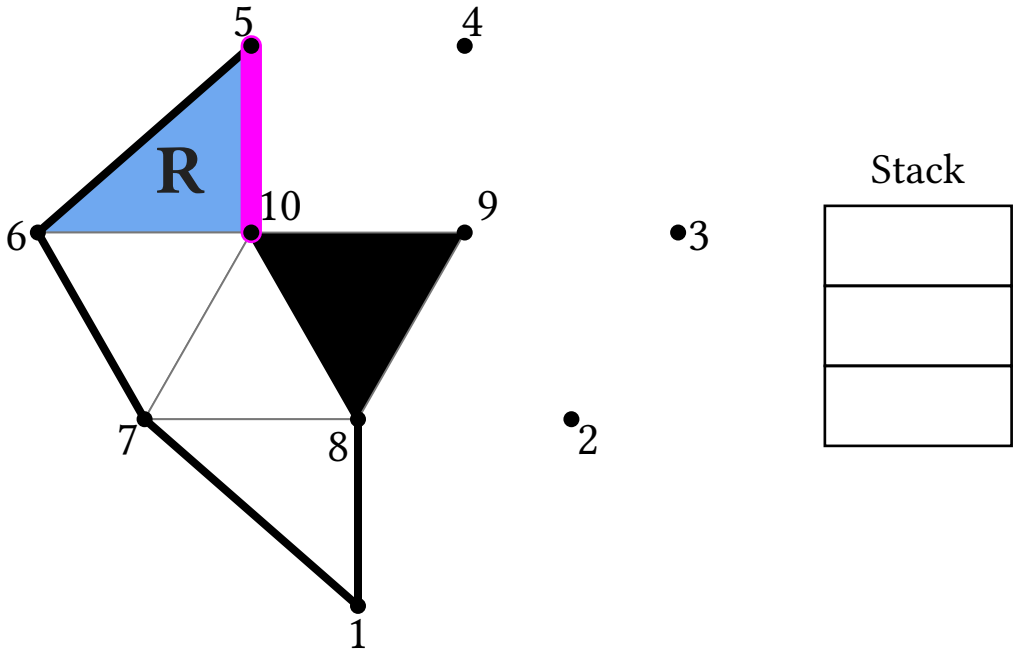


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRR

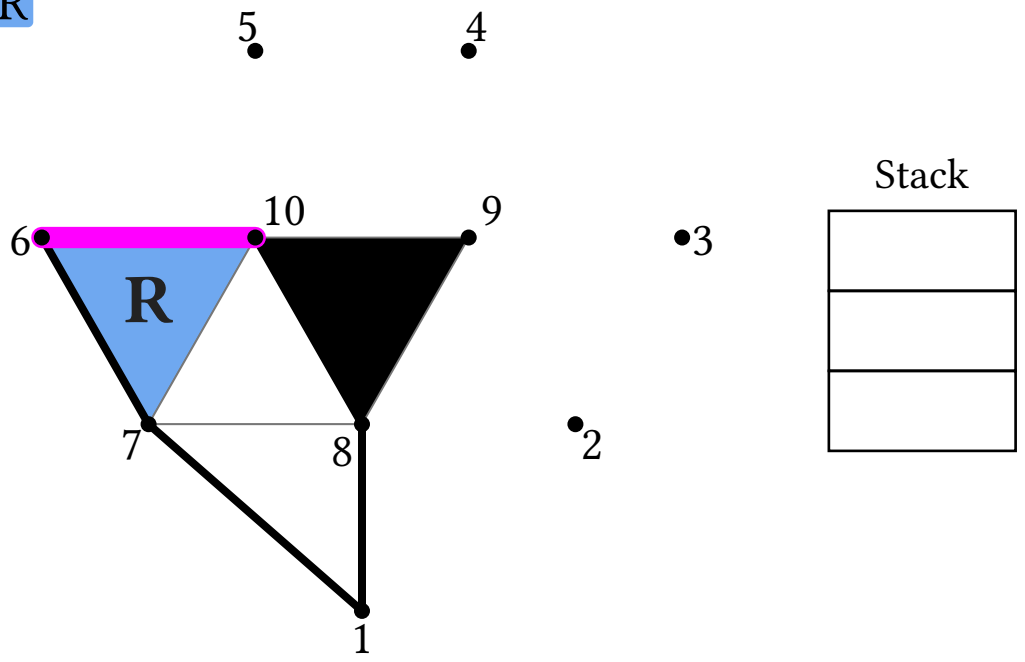


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRL

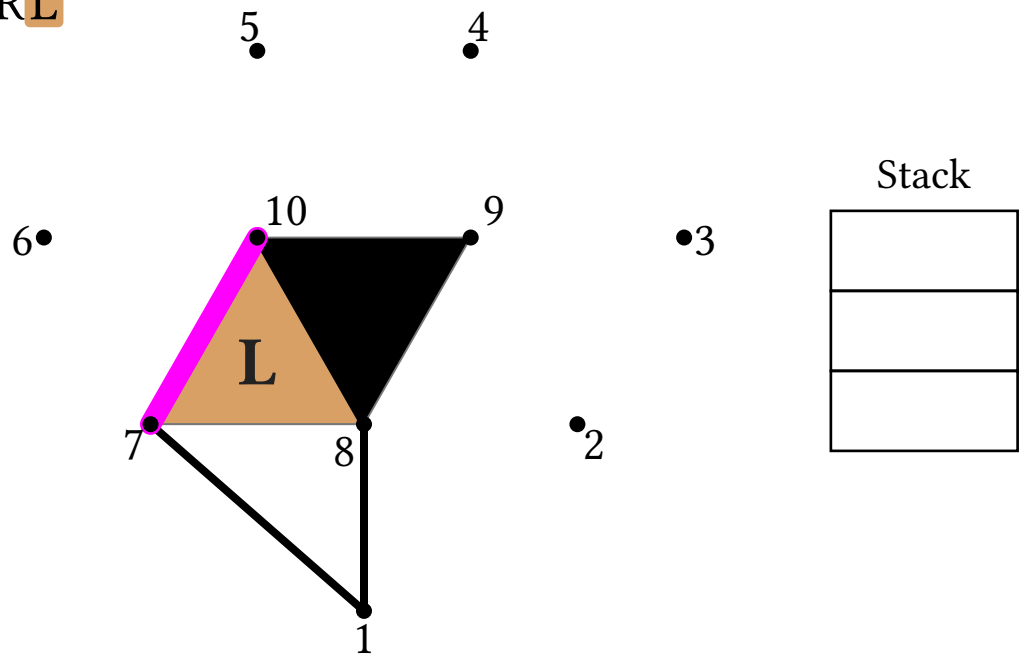


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

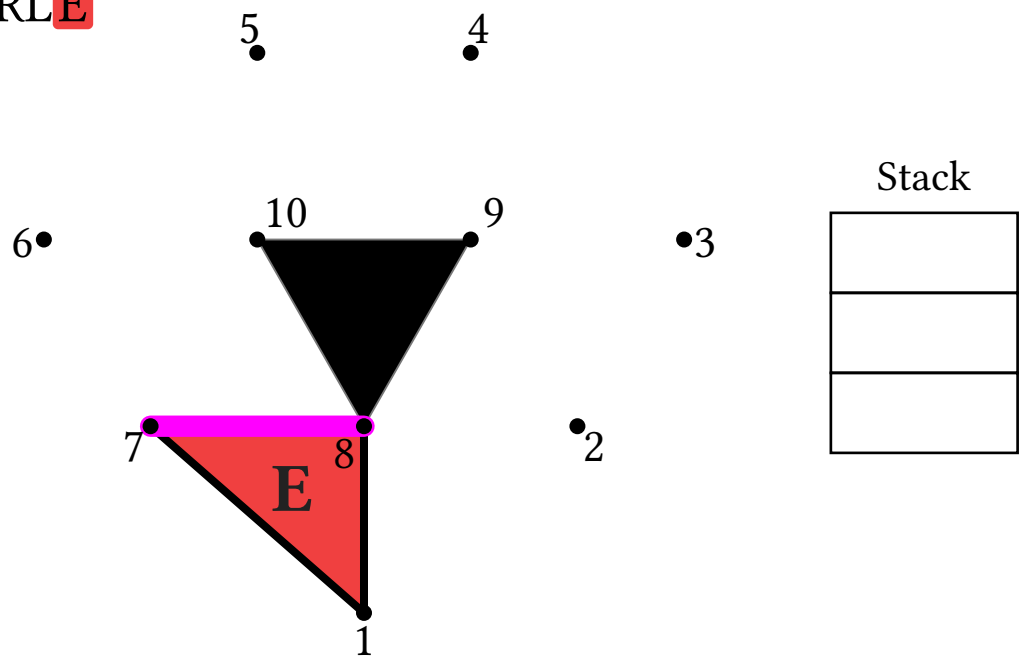


Figure 7: Komprese díry.

3.3 Vyšší genus?

History = SLRRRLRRLE

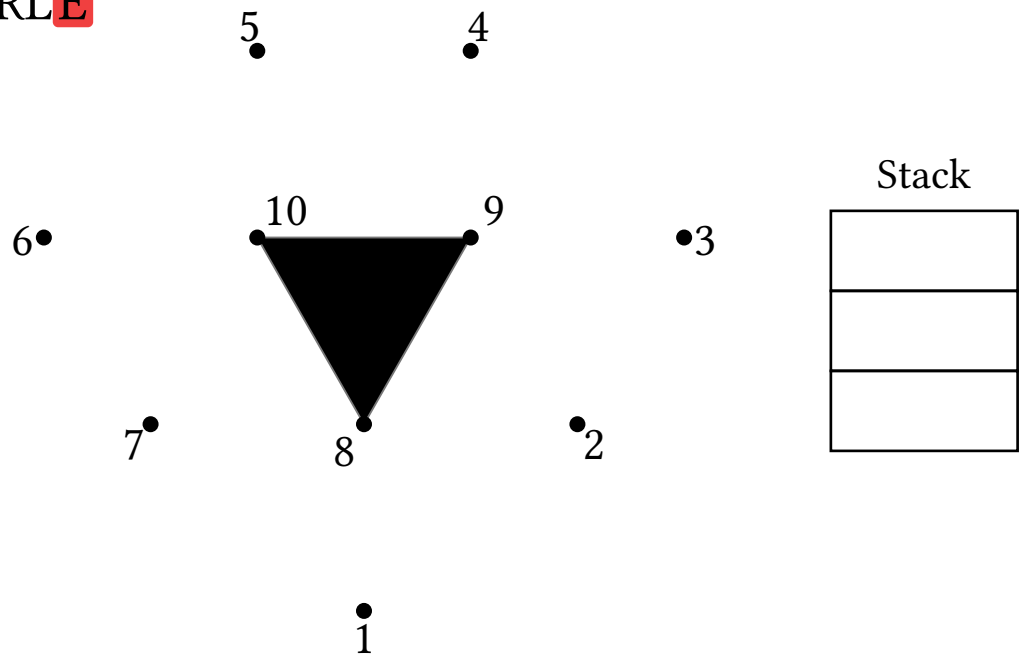


Figure 7: Komprese díry.

4. Dekomprese

4.1 Možnosti

4.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

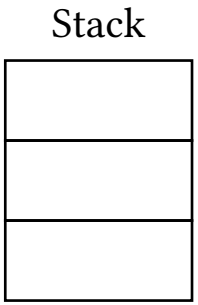


Figure 8: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

Stack

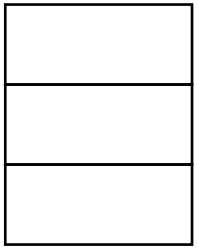
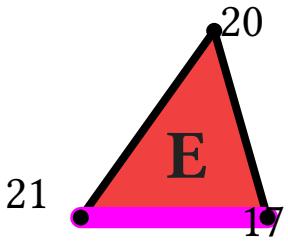


Figure 8: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



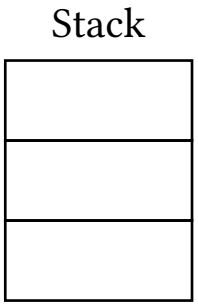
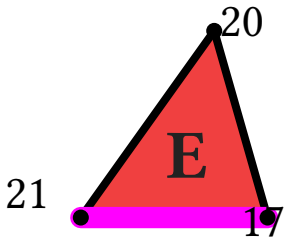


Figure 8: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



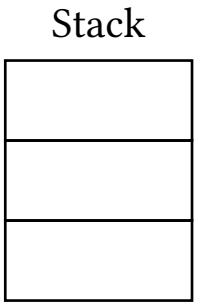


Figure 8: Dekomprese.

5. Zdroje

5.1 Seznam figures

Figure 1: Dobový kontext vzniku algoritmu Edgebreaker. 2
Figure 2: Jarek Rossignac, zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg 3
Figure 3: Trojúhelníková mesh. 5
Figure 4: Jednoduchá komprese. 9
Figure 5: Komprese díry. 10
Figure 6: Dekomprese. 14

5.2 Obrázky (dobový kontext)

1	**Nagano** (2/1998)	Wikipedia, *fair use*
2	**Duda** (4/1998)	Gov.pl, CC BY 3.0 PL, Wikimedia Commons
3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
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5.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

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